Supplementary Material

Differentiable Interval Bottlenecks for Interpretable Anomaly Detection in Numerical Data

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***Abstract*—This supplement is the complete experimental record behind the main paper, together with the material deferred there for space. It is organized as a roadmap: Section I details the datasets, baselines, and evaluation protocol; Section II the per-dataset robustness ablations; Section III the empirical certification of the clipping-margin theorem; Section IV the interpretability evaluation suite (label-free ranking, faithfulness perturbation, conciseness, and human-readability); Section V the complexity and scalability profile; and Section VI the *complete* per-dataset ROC–AUC and AUPRC for all 48 ADBench datasets and 25 methods (mean ± std over 10 seeds, with ranks and top-3 highlighting) plus the per-scale critical-difference diagrams. All proofs are in the appendix of the main paper. Code and this document: https://github.com/DiffInt/diffint.**

I. Datasets and experimental protocol

A. Datasets

We evaluate on the 48 numerical datasets of the ADBench benchmark, grouped by scale into *small* (12), *medium* (15), *large* (11), and *high-dimensional* (10). Tables I–IV list, per dataset, the number of samples, features, and outliers, the outlier percentage, and the batch size used by DiffInt (64 for $n \leq$ 10k, 512 for 10k $< n \leq$ 20k, 1024 otherwise). The suite spans four orders of magnitude in sample size ($n \in [80,\ 6.2\times10^5]$) and from 3 to 1,555 features, so it stresses both data-scarce and high-dimensional regimes.

TABLE I: Small-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
4_breastw	683	9	239	34.99	64
14_glass	214	7	9	4.21	64
15_Hepatitis	80	19	13	16.25	64
18_Ionosphere	351	33	126	35.90	64
21_Lymphography	148	18	6	4.05	64
29_Pima	768	8	268	34.90	64
37_Stamps	340	9	31	9.12	64
39_vertebral	240	6	30	12.50	64
42_WBC	223	9	10	4.48	64
43_WDBC	367	30	10	2.72	64
45_wine	129	13	10	7.75	64
46_WPBC	198	33	47	23.74	64

TABLE II: Medium-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
2_annthyroid	7,200	6	534	7.42	64
6_cardio	1,831	21	176	9.61	64
7_Cardiotocography	2,114	21	466	22.04	64
12_fault	1,941	27	673	34.67	64
19_landsat	6,435	36	1,333	20.71	64
20_letter	1,600	32	100	6.25	64
27_PageBlocks	5,393	10	510	9.46	64
28_pendigits	6,870	16	156	2.27	64
30_satellite	6,435	36	2,036	31.64	64
31_satimage-2	5,803	36	71	1.22	64
38_thyroid	3,772	6	93	2.47	64
40_vowels	1,456	12	50	3.43	64
41_Waveform	3,443	21	100	2.90	64
44_Wilt	4,819	5	257	5.33	64
47_yeast	1,500	8	20	1.33	64

TABLE III: Large-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
1_ALOI	49,534	27	1,508	3.04	1,024
8_celeba	202,599	39	4,547	2.24	1,024
10_cover	286,048	10	2,747	0.96	1,024
11_donors	619,326	10	36,710	5.93	1,024
13_fraud	284,807	29	492	0.17	1,024
16_http	567,498	3	2,211	0.39	1,024
22_magic.gamma	19,020	10	6,688	35.16	512
23_mammography	11,183	6	260	2.32	512
32_shuttle	49,097	9	3,511	7.15	1,024
33_skin	245,057	3	50,859	20.75	1,024
34_smtp	95,156	3	30	0.03	1,024

TABLE IV: High-dimensional datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
3_backdoor	95,329	196	2,329	2.44	1,024
5_campaign	41,188	62	4,640	11.27	1,024
9_census	299,285	500	18,568	6.20	1,024
17_InternetAds	1,966	1,555	368	18.72	64
24_mnist	7,603	100	700	9.21	64
25_musk	3,062	166	97	3.17	64
26_optdigits	5,216	64	150	2.88	64
35_SpamBase	4,207	57	1,679	39.91	64
36_speech	3,686	400	61	1.65	64
48_arrhythmia	452	274	66	14.60	64

B. Protocol, baselines, and metrics

Protocol. DiffInt is trained *semi-supervised* on inliers only (label 0), with a 40% test split and 10 random seeds, under a single $[-1,1]$-normalized configuration shared across all datasets (lr 5×10^{-5}, K=200, τ=0.1, ρ=0.999, 1000 epochs, Adam); no hyper-parameter is tuned per dataset. **Baselines.** We compare against 22 baselines from the ADBench/TCCM codebase: classical detectors (ABOD, LOF, PCA, KPCA, OCSVM, ECOD, CBLOF, KDE, GMM, MCD, HBOS, IForest, LMDD, QMCD, Sampling) and deep/inductive ones (AutoEncoder, VAE, DIF, LUNAR, SO/MO_GAAL, and the flow-matching model TCCM); with the three DiffInt variants (PA/IA/DA)

TABLE V: Input normalization. Mapping into a bounded symmetric domain is what matters; among bounded/standardized choices $[-1, 1]$ is the most accurate, within 1.5 points of the others.

Dataset	none	$[0, 1]$	standard	$[-1, 1]$
45_wine	**0.990**	0.932	0.883	0.950
21_Lymphography	0.990	**0.998**	0.996	0.994
46_WPBC	0.510	0.543	0.541	**0.584**
14_glass	**0.896**	0.886	0.877	0.876
42_WBC	0.990	0.989	0.990	**0.998**
39_vertebral	0.335	0.498	**0.511**	0.477
37_Stamps	0.924	0.921	0.929	**0.953**
Mean	0.805	0.824	0.818	**0.833**

TABLE VI: Training contamination: ROC–AUC vs. fraction of anomalies injected into the "inlier" training set. Degradation is graceful.

Dataset	0%	5%	10%	20%
45_wine	0.950	0.659	0.645	0.624
21_Lymphography	0.994	0.978	0.970	0.973
46_WPBC	0.584	0.565	0.570	0.520
14_glass	0.876	0.755	0.728	0.726
42_WBC	0.998	0.979	0.982	0.976
39_vertebral	0.477	0.458	0.432	0.456
37_Stamps	0.953	0.865	0.816	0.766
Mean	0.833	0.751	0.735	0.720

this gives the 25 methods ranked in the tables. All methods share the same $[-1, 1]$ normalization. **Metrics.** We report ROC–AUC (area under the ROC curve) and AUPRC (average precision); both are threshold-free and higher-is-better. **Missing runs.** A few baselines did not complete on the largest datasets within budget; each such run is floored to the worst rank (the number of methods) on that dataset, the standard conservative convention used in the main paper. **How to read the result tables.** Every cell is mean$_{\pm\text{std}}$(rank) over the seeds; per column we highlight **rank 1**, rank 2, and rank 3.

II. Robustness Ablations

We expand the main-paper ablations to per-dataset numbers on seven small datasets ($n \geq 100$; PA, K=200, 3 seeds; ROC–AUC), probing two design choices; Hepatitis (n=80) is excluded as too small for a stable inlier-only test split. **Normalization** (Table V): mapping into a bounded symmetric domain is what matters. Dropping it ("none") is the only choice that collapses on several datasets, while among bounded/standardized choices $[-1, 1]$ is the most accurate, within 1.5 points of $[0, 1]$ and StandardScaler; we adopt it because it makes the theory transfer (main paper, Sec. III-B) and it is also the top performer. **Training contamination** (Table VI): injecting anomalies into the "inlier" training set degrades accuracy *gracefully* rather than catastrophically, since the EMA support averages over batches so a minority of contaminating points cannot dominate the learned intervals.

a) Effect of bottleneck width K: selective reconstruction.: Companion to Fig. 4 of the main paper. As K grows, the

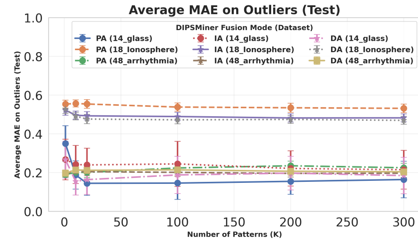


Fig. 1: Outlier reconstruction MAE vs. bottleneck width K (per dataset). The error on outliers remains high as K grows, evidence of selective reconstruction; K is capacity, not a pattern-set size.

TABLE VII: Certification of the clipping margin (Certified-Lipschitz DiffInt-PA, K=200, 3 seeds). AUROC for the unconstrained (default) and certified decoders; L: Lipschitz upper bound; *collapse*: mean distance of out-of-support reconstructions to $g_\theta(f_0)$ (fraction of inlier amplitude); *bound*: fraction of out-of-support points satisfying the inequality.

	AUROC				
Dataset	unc.	cert.	L	collapse	bound
14_glass	0.891	0.744	1.05	0.52	100%
42_WBC	0.930	0.903	1.03	0.29	100%
18_Ionosphere	0.953	0.828	1.04	0.24	100%
29_Pima	0.696	0.686	1.04	0.53	100%
6_cardio	0.976	0.955	1.06	0.45	100%
31_satimage-2	0.999	0.987	1.07	0.38	100%
Mean	0.908	0.851	1.05	0.40	100%

reconstruction error on *outliers* stays high (Fig. 1) while inlier error drops: the model reconstructs the inlier manifold faithfully but keeps out-of-support points far from their input, the empirical counterpart of the clipping margin (Thm. 1). Accuracy therefore saturates after a few units and we fix K=200 as a capacity choice.

III. Certifying the Clipping Margin

We test Theorem 1 of the main paper with the Certified-Lipschitz DiffInt-PA (spectrally normalized decoder, no non-1-Lipschitz layers). On the low-membership test points (inside no learned box) we measure: the decoder Lipschitz upper bound $L = \prod_l \|W_l\|_2$; the fraction of those points satisfying the certified inequality $\mathrm{MAE}(x) \geq \frac{1}{d}\|x - g_\theta(f_0)\|_1 - \frac{L}{\sqrt{d}}\|\phi(x) - \phi_0\|_2$; and the collapse of their reconstructions toward the empty-code image $g_\theta(f_0)$ (mean distance as a fraction of inlier amplitude). The bound is respected for every out-of-support point, $L \approx 1$ is an explicit constant, and the collapse is tight. Certification trades some accuracy, so the unconstrained decoder remains the default (Table VII).

a) Checklist for the conditions.: The hypotheses of Theorem 1 are operational. Coordinate j is *active* for unit k iff its mean inlier membership is bounded away from 1 ($\bar{I}_{kj} < 0.9$), i.e. it constrains inliers there. A point is *out-of-support by margin η* when its box membership drops below $\beta = \sigma(-\eta/\tau)$ on those coordinates (η=0.2, β≈0.12). Under this check, $\geq 99\%$

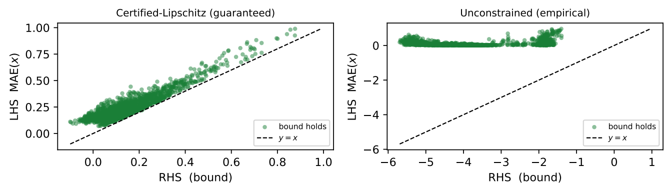


Fig. 2: Theorem-to-practice bridge: realized error LHS $=$ $\mathrm{MAE}(x)$ vs. the clipping-margin lower bound RHS, for out-of-support points (pooled over glass, ionosphere, cardio; PA). *Left*: Certified-Lipschitz decoder, the bound is tight and respected by every point (0 violations). *Right*: unconstrained decoder, the inequality still holds but the bound is loose and not certified (note the differing x-scale), an empirical phenomenon rather than a guarantee.

of test anomalies are out-of-support (glass/ionosphere/cardio), so the clipping regime applies broadly; the strict full-collapse hypothesis (every unit) is the extreme covered by the graded suppression of Cor. 1.

b) Bridge: bound vs. realized error.: Fig. 2 plots the two sides of the bound for out-of-support points. For the Certified-Lipschitz variant every point lies above $y = x$ (LHS $\geq$ RHS, no violations) and the bound is tight. The unconstrained variant satisfies the same inequality empirically, but the bound is loose and uncertified (note the x-scale; L is not a verified Lipschitz constant there), consistent with reading it as an empirical mechanism (main paper, Remark 1).

IV. Interpretability as Evidence

We support the two interpretability claims of the main paper with quantitative measurements (PA, K=200, 3 seeds; Table VIII).

(1) Label-free ranking. For deployment we rank (unit, feature) constraints by a label-free importance $\mathrm{LFI}_{kj} = s_{kj} \cdot (1 - \Delta_{kj}/\mathrm{range}_j) \cdot \mathrm{stab}_{kj}$ (support $\times$ tightness $\times$ stability), using only quantities the model already maintains. It agrees with the label-based ICR importance (Spearman ρ; mean 0.61), so the same constraints can be surfaced when labels are unavailable.

(2) Faithfulness perturbation. For each test inlier we perturb its top-LFI features to *cross* their interval boundaries and measure the anomaly-score rise Δ_{top}, against perturbing the same number of random features Δ_{rand}. The top-ranked constraints raise the score $1.5\times$ more on average (on 69% of inliers), so the highlighted constraints *causally* drive the score rather than merely correlating with it.

(3) Conciseness and readability. For a flagged point, the per-feature reconstruction error concentrates: on average ≈half the features (10.8 of 20; Table IX) account for 80% of the anomaly score, so a short shortlist suffices to explain it. The interval units themselves are *soft and wide* (mean width 0.70 of the feature range, and a unit softly constrains most coordinates by the membership criterion of Def. 1), which is precisely why we present intervals as *candidate constraints* rather than crisp rules and surface only the top-k by LFI ($k = 2$ in the main-paper read-out) rather than a whole unit.

TABLE VIII: Interpretability as evidence. Spearman ρ between the label-free ranking (LFI) and the label-based ICR importance; and a perturbation test: score rise from crossing the top-LFI feature boundaries (Δ_{top}) vs. random features (Δ_{rand}), their ratio, and the fraction of inliers with $\Delta_{\text{top}} > \Delta_{\text{rand}}$.

Dataset	ρ(LFI,ICR)	Δ_{top}	Δ_{rand}	ratio	% top>rand
14_glass	0.357	0.200	0.134	1.48	71%
42_WBC	0.487	0.094	0.074	1.26	62%
18_Ionosphere	0.645	0.105	0.078	1.34	64%
29_Pima	0.789	0.171	0.100	1.70	71%
6_cardio	0.762	0.119	0.066	1.80	79%
Mean	0.608	0.138	0.091	1.52	69%

TABLE IX: Conciseness and human-readability proxies (PA, K=200, 3 seeds). #feat$_{80\%}$: features (ranked by per-feature error) explaining 80% of a flagged point's score; active/unit: mean number of active features ($\bar{I}_{kj} < 0.9$) per top interval unit; width: mean active-interval width as a fraction of the $[-1, 1]$ range.

Dataset	d	#feat$_{80\%}$	active/unit	width
14_glass	9	4.8	6.7	0.70
42_WBC	30	16.0	24.8	0.69
18_Ionosphere	33	17.9	29.8	0.70
29_Pima	8	4.6	5.3	0.70
6_cardio	21	11.0	17.2	0.70
Mean	20	10.8	16.8	0.70

V. Complexity and Scalability

Practical deployability matters as much as accuracy, so we summarize the resource profile of DiffInt. Let n be the number of training points, d the feature count, K the number of interval units, B the batch size, E the epochs, and H the (fixed) decoder hidden width; the decoder is an MLP whose size is independent of n.

Parameter count. The interval bottleneck stores a centre and a softplus half-width per (unit, feature): $2Kd$ parameters, linear in K and d and *independent of* n. The decoder adds a compact MLP ($O(KH + Hd)$ weights), and IA/DA add a *single* per-feature value network *shared* across all d coordinates (a fixed cost, not $O(d)$). The total stays MLP-class (10^5–10^6 weights even at d=1,555, InternetAds), far below typical deep tabular detectors. Table X reports the $2Kd$ bottleneck size and the matched-budget training time across scales.

Training complexity. A minibatch computes Bd sigmoid pairs for the soft memberships, an $O(BKd)$ aggregation (a per-unit log-sum over features plus an $O(K)$ softmax for PA), and one decoder forward/backward; an epoch is thus $O(nKd)$ plus the n-independent decoder cost, with *no* pairwise, kernel, or neighbour computation. Total training is $O(E\,nKd)$.

Runtime vs. baselines. DiffInt uses one forward pass per sample, unlike memory-based detectors (e.g. LUNAR) whose kNN queries grow with n. Empirically (main paper, Table II,

TABLE X: Resource profile across scales (K=200). Bottleneck parameters ($2Kd$) are linear in d and independent of n; training time (DIFFINT-PA, matched 1000-epoch budget) is reproduced from Table II of the main paper and grows with n and d.

Dataset	n	d	bottleneck $2Kd$	train (s)
6_cardio	1,831	21	8,400	3.0
31_satimage-2	5,803	36	14,400	54.6
26_optdigits	5,216	64	25,600	60.1
48_arrhythmia	452	274	109,600	30.2
32_shuttle	49,097	9	3,600	193.5

matched 1000-epoch budget) DIFFINT-PA trains one to two orders of magnitude faster than the binary differentiable miner BinaPs, faster than LUNAR on most datasets, and scales to the largest ADBench sets ($n \approx 6.2 \times 10^5$); PA is the cheapest variant, IA/DA cost more through the value network.

Inference and memory footprint. Scoring a point is a single forward pass: $O(Kd)$ memberships, the aggregation, the decoder, then the MAE. DIFFINT is *inductive*—it keeps only the trained parameters, not the training set—so inference latency and memory are constant in n, whereas transductive/memory-based detectors must retain or query the data. The training footprint is the parameters ($O(Kd)$), one batch of memberships ($O(BKd)$ activations), and the $K \times d$ EMA support buffer ($O(Kd)$); none depends on n. At inference only the $O(Kd)$ parameters and a single $K \times d$ membership tensor are held.

The K knob. K trades capacity for cost linearly. ROC–AUC saturates after a few units (main paper, Fig. 4), so K can be reduced well below the default under tight budgets with little accuracy loss; we keep K=200 as one untuned default across all 48 datasets.

VI. Complete detection accuracy

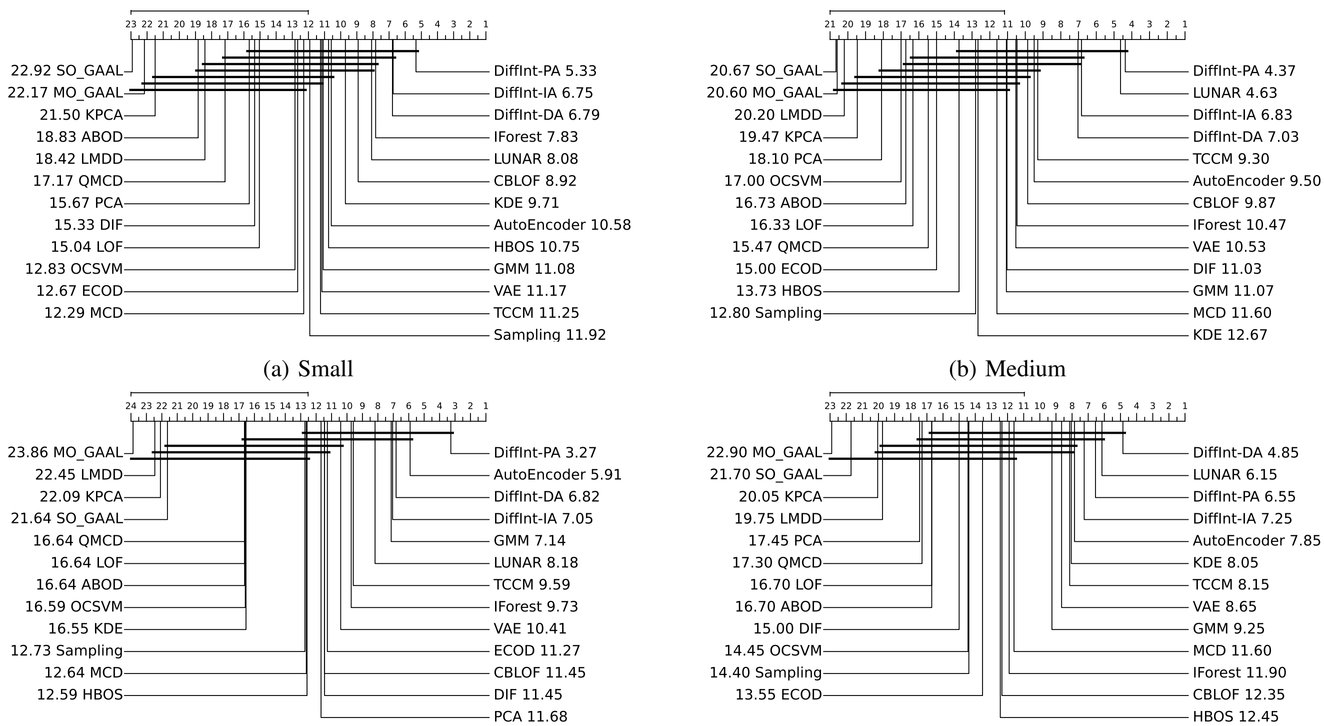
This section reports the full per-dataset detection accuracy underlying the critical-difference analysis of the main paper: ROC–AUC (Sec. VI-A) and average precision (Sec. VI-B). Across both metrics the three DiffInt variants sit in the leading group on every scale, with DiffInt-PA most often first on small/medium/large data and DiffInt-DA strongest in high dimensions.

A. ROC–AUC

Tables XI–XIX give the per-dataset ROC–AUC. Fig. 3 shows the per-scale critical-difference diagrams (the overall diagram is in the main paper); they confirm that the DiffInt cluster is statistically ahead of the classical and adversarial baselines, most clearly on small and large data.

B. Average precision (AUPRC)

Tables XX–XXVIII give the per-dataset AUPRC under the same protocol and highlighting. AUPRC weights the rare anomaly class more heavily than ROC–AUC; the DiffInt ranking is consistent across the two metrics, indicating the gains are not an artifact of the chosen measure.



(a) Small

(b) Medium

(c) Large

(d) High-dimensional

Fig. 3: ROC–AUC critical-difference diagrams per dataset scale (missing floored to worst rank). DIFFINT-PA leads on small/medium/large; DIFFINT-DA leads on high-dimensional data, with LUNAR next; bars indicate non-significant differences.

TABLE XI: ROC–AUC, small datasets (1/2).

Model	4_breastw	14_glass	15_Hepatitis	18_Ionosphere	21_Lymphography	29_Pima
ABOD	0.3737$_{\pm 0.0338}$(22)	0.8264$_{\pm 0.0697}$(10)	0.6542$_{\pm 0.1520}$(21)	0.9255$_{\pm 0.0185}$(13)	0.9682$_{\pm 0.0240}$(20)	0.6548$_{\pm 0.0408}$(16)
AutoEncoder	0.9855$_{\pm 0.0075}$(14)	0.7107$_{\pm 0.0779}$(18)	0.8501$_{\pm 0.0387}$(2)	0.9604$_{\pm 0.0090}$(6)	0.9904$_{\pm 0.0087}$(14)	0.7070$_{\pm 0.0239}$(10)
CBLOF	0.9939$_{\pm 0.0026}$(5)	0.8784$_{\pm 0.0345}$(5)	0.8089$_{\pm 0.0874}$(11)	0.9557$_{\pm 0.0223}$(9)	0.9816$_{\pm 0.0438}$(17)	0.7006$_{\pm 0.0232}$(11)
DIF	0.8377$_{\pm 0.0375}$(18)	0.8562$_{\pm 0.0371}$(7)	0.7723$_{\pm 0.0901}$(18)	0.9587$_{\pm 0.0074}$(8)	0.8833$_{\pm 0.0992}$(21)	0.6144$_{\pm 0.0258}$(21)
DiffInt-DA	0.9925$_{\pm 0.0027}$(8)	0.9056$_{\pm 0.0202}$(2)	0.7824$_{\pm 0.0339}$(16)	0.9602$_{\pm 0.0065}$(7)	0.9988$_{\pm 0.0036}$(8)	0.6684$_{\pm 0.0149}$(15)
DiffInt-IA	0.9896$_{\pm 0.0030}$(12)	0.9166$_{\pm 0.0224}$(1)	0.8153$_{\pm 0.0284}$(9)	0.9609$_{\pm 0.0073}$(5)	0.9956$_{\pm 0.0046}$(11)	0.6775$_{\pm 0.0145}$(13)
DiffInt-PA	0.9954$_{\pm 0.0018}$(1)	0.8788$_{\pm 0.0245}$(4)	0.8333$_{\pm 0.0401}$(7)	0.9757$_{\pm 0.0030}$(2)	0.9997$_{\pm 0.0014}$(2)	0.7304$_{\pm 0.0113}$(3)
ECOD	0.9932$_{\pm 0.0029}$(7)	0.6911$_{\pm 0.0797}$(19)	0.8402$_{\pm 0.0550}$(3)	0.7745$_{\pm 0.0221}$(19)	0.9990$_{\pm 0.0022}$(4)	0.6374$_{\pm 0.0239}$(18)
GMM	0.9863$_{\pm 0.0055}$(13)	0.7508$_{\pm 0.0624}$(16)	0.8018$_{\pm 0.0785}$(14)	0.9664$_{\pm 0.0071}$(3)	0.9873$_{\pm 0.0092}$(15)	0.7202$_{\pm 0.0217}$(7)
HBOS	0.9911$_{\pm 0.0036}$(9)	0.7848$_{\pm 0.0621}$(14)	0.8342$_{\pm 0.0407}$(6)	0.7334$_{\pm 0.0519}$(20)	0.9975$_{\pm 0.0045}$(9)	0.7339$_{\pm 0.0201}$(2)
IForest	0.9942$_{\pm 0.0028}$(4)	0.7855$_{\pm 0.0429}$(13)	0.8348$_{\pm 0.0613}$(5)	0.9116$_{\pm 0.0189}$(15)	0.9991$_{\pm 0.0019}$(3)	0.7272$_{\pm 0.0275}$(5)
KDE	0.9946$_{\pm 0.0025}$(2)	0.7622$_{\pm 0.0454}$(15)	0.7942$_{\pm 0.0540}$(15)	0.9520$_{\pm 0.0105}$(10)	0.9988$_{\pm 0.0025}$(6)	0.7208$_{\pm 0.0226}$(6)
KPCA	0.1983$_{\pm 0.0825}$(23)	0.3975$_{\pm 0.1372}$(22)	0.4379$_{\pm 0.1895}$(23)	0.4606$_{\pm 0.0632}$(23)	0.4137$_{\pm 0.2537}$(22)	0.4976$_{\pm 0.0274}$(23)
LMDD	0.7050$_{\pm 0.1609}$(19)	0.3743$_{\pm 0.1793}$(24)	0.5742$_{\pm 0.1757}$(22)	0.6665$_{\pm 0.1000}$(21)	0.9759$_{\pm 0.0186}$(19)	0.6293$_{\pm 0.0860}$(20)
LOF	0.4482$_{\pm 0.0215}$(21)	0.8048$_{\pm 0.0784}$(11)	0.8211$_{\pm 0.0761}$(8)	0.8665$_{\pm 0.0479}$(16)	0.9988$_{\pm 0.0025}$(6)	0.6348$_{\pm 0.0227}$(19)
LUNAR	0.9902$_{\pm 0.0046}$(11)	0.8816$_{\pm 0.0359}$(3)	0.7782$_{\pm 0.0579}$(17)	0.9781$_{\pm 0.0085}$(1)	0.9906$_{\pm 0.0112}$(13)	0.7296$_{\pm 0.0286}$(4)
MCD	0.9850$_{\pm 0.0084}$(15)	0.7869$_{\pm 0.0347}$(12)	0.8088$_{\pm 0.0621}$(13)	0.9616$_{\pm 0.0092}$(4)	0.9861$_{\pm 0.0140}$(16)	0.7158$_{\pm 0.0222}$(8)
MO_GAAL	0.0117$_{\pm 0.0142}$(24)	0.3744$_{\pm 0.2277}$(23)	0.3695$_{\pm 0.1895}$(24)	0.3329$_{\pm 0.0612}$(24)	0.0085$_{\pm 0.0148}$(24)	0.3319$_{\pm 0.0499}$(25)
OCSVM	0.9945$_{\pm 0.0025}$(3)	0.4470$_{\pm 0.1473}$(21)	0.8088$_{\pm 0.0630}$(12)	0.8405$_{\pm 0.0192}$(17)	0.9988$_{\pm 0.0025}$(6)	0.6797$_{\pm 0.0246}$(12)
PCA	0.9495$_{\pm 0.0188}$(17)	0.6551$_{\pm 0.0838}$(20)	0.8120$_{\pm 0.0635}$(10)	0.7864$_{\pm 0.0215}$(18)	0.9962$_{\pm 0.0058}$(10)	0.6377$_{\pm 0.0241}$(17)
QMCD	0.4800$_{\pm 0.1746}$(20)	0.8423$_{\pm 0.0490}$(8)	0.7150$_{\pm 0.1246}$(20)	0.5509$_{\pm 0.0455}$(22)	0.0884$_{\pm 0.1204}$(23)	0.7417$_{\pm 0.0239}$(1)
SO_GAAL	0.0110$_{\pm 0.0116}$(25)	0.3392$_{\pm 0.1817}$(25)	0.3134$_{\pm 0.1751}$(25)	0.2823$_{\pm 0.0611}$(25)	0.0049$_{\pm 0.0108}$(25)	0.3350$_{\pm 0.0491}$(24)
Sampling	0.9934$_{\pm 0.0033}$(6)	0.8696$_{\pm 0.0480}$(6)	0.7645$_{\pm 0.0976}$(19)	0.9395$_{\pm 0.0256}$(12)	0.9780$_{\pm 0.0359}$(18)	0.7115$_{\pm 0.0369}$(9)
TCCM	0.9677$_{\pm 0.0195}$(16)	0.8334$_{\pm 0.0914}$(9)	0.8394$_{\pm 0.0708}$(4)	0.9400$_{\pm 0.0158}$(11)	1.0000$_{\pm 0.0000}$(1)	0.5562$_{\pm 0.1001}$(22)
VAE	0.9903$_{\pm 0.0045}$(10)	0.7148$_{\pm 0.0655}$(17)	0.8600$_{\pm 0.0425}$(1)	0.9206$_{\pm 0.0155}$(14)	0.9932$_{\pm 0.0068}$(12)	0.6714$_{\pm 0.0323}$(14)

TABLE XII: ROC–AUC, small datasets (2/2).

Model	37_Stamps	39_vertebral	42_WBC	43_WDBC	45_wine	46_WPBC
ABOD	0.6954$_{\pm0.0525}$(21)	0.4214$_{\pm0.0847}$(20)	0.9297$_{\pm0.0301}$(20)	0.9143$_{\pm0.0504}$(20)	0.5828$_{\pm0.2278}$(22)	0.4897$_{\pm0.0563}$(21)
AutoEncoder	0.9217$_{\pm0.0398}$(12)	0.4644$_{\pm0.1041}$(14)	0.9841$_{\pm0.0220}$(14)	0.9937$_{\pm0.0055}$(4)	0.9364$_{\pm0.0499}$(6)	0.5178$_{\pm0.0481}$(13)
CBLOF	0.9457$_{\pm0.0235}$(6)	0.4711$_{\pm0.0742}$(12)	0.9961$_{\pm0.0039}$(2)	0.9860$_{\pm0.0099}$(16)	0.9517$_{\pm0.0376}$(2)	0.5280$_{\pm0.0407}$(11)
DIF	0.9426$_{\pm0.0215}$(7)	0.5781$_{\pm0.0895}$(3)	0.7821$_{\pm0.0531}$(21)	0.7227$_{\pm0.0519}$(21)	0.7978$_{\pm0.1078}$(19)	0.4971$_{\pm0.0485}$(20)
DiffInt-DA	0.9569$_{\pm0.0152}$(2)	0.5706$_{\pm0.0459}$(4)	0.9918$_{\pm0.0061}$(8)	0.9994$_{\pm0.0012}$(2)	0.9325$_{\pm0.0334}$(10)	0.5702$_{\pm0.0196}$(**1**)
DiffInt-IA	0.9587$_{\pm0.0132}$(**1**)	0.5618$_{\pm0.0444}$(5)	0.9910$_{\pm0.0102}$(9)	0.9998$_{\pm0.0010}$(**1**)	0.9339$_{\pm0.0319}$(7)	0.5399$_{\pm0.0265}$(7)
DiffInt-PA	0.9516$_{\pm0.0116}$(3)	0.5274$_{\pm0.0429}$(6)	0.9973$_{\pm0.0050}$(**1**)	0.9741$_{\pm0.0076}$(18)	0.8852$_{\pm0.0594}$(15)	0.5652$_{\pm0.0222}$(2)
ECOD	0.9157$_{\pm0.0244}$(15)	0.4645$_{\pm0.0796}$(13)	0.9953$_{\pm0.0046}$(6)	0.9870$_{\pm0.0068}$(13)	0.8484$_{\pm0.0960}$(16)	0.5029$_{\pm0.0422}$(19)
GMM	0.9270$_{\pm0.0226}$(11)	0.4909$_{\pm0.0550}$(10)	0.9767$_{\pm0.0198}$(18)	0.9882$_{\pm0.0093}$(11)	0.9677$_{\pm0.0268}$(**1**)	0.5141$_{\pm0.0507}$(14)
HBOS	0.9375$_{\pm0.0146}$(9)	0.3982$_{\pm0.0437}$(23)	0.9896$_{\pm0.0093}$(11)	0.9890$_{\pm0.0063}$(9)	0.9296$_{\pm0.0419}$(11)	0.5483$_{\pm0.0532}$(6)
IForest	0.9485$_{\pm0.0202}$(4)	0.4426$_{\pm0.1017}$(17)	0.9948$_{\pm0.0049}$(7)	0.9903$_{\pm0.0056}$(6)	0.9393$_{\pm0.0441}$(5)	0.5287$_{\pm0.0451}$(10)
KDE	0.9309$_{\pm0.0174}$(10)	0.4096$_{\pm0.0856}$(22)	0.9960$_{\pm0.0045}$(4)	0.9864$_{\pm0.0075}$(15)	0.9336$_{\pm0.0339}$(8)	0.5603$_{\pm0.0511}$(3)
KPCA	0.4111$_{\pm0.1136}$(23)	0.5204$_{\pm0.0619}$(8)	0.3710$_{\pm0.2036}$(22)	0.2678$_{\pm0.0890}$(23)	0.3955$_{\pm0.2631}$(23)	0.4644$_{\pm0.0961}$(23)
LMDD	0.4454$_{\pm0.0266}$(22)	0.4355$_{\pm0.0811}$(18)	0.9769$_{\pm0.0615}$(17)	0.9941$_{\pm0.0033}$(3)	0.6987$_{\pm0.1469}$(20)	0.5125$_{\pm0.0619}$(16)
LOF	0.7229$_{\pm0.1204}$(20)	0.3515$_{\pm0.0717}$(25)	0.9759$_{\pm0.0198}$(19)	0.9871$_{\pm0.0088}$(12)	0.9115$_{\pm0.0592}$(14)	0.5326$_{\pm0.0635}$(9)
LUNAR	0.9473$_{\pm0.0248}$(5)	0.4484$_{\pm0.0817}$(15)	0.9794$_{\pm0.0237}$(16)	0.9935$_{\pm0.0055}$(5)	0.9478$_{\pm0.0439}$(3)	0.5536$_{\pm0.0417}$(4)
MCD	0.8368$_{\pm0.0255}$(19)	0.4872$_{\pm0.0529}$(11)	0.9807$_{\pm0.0230}$(15)	0.9682$_{\pm0.0141}$(19)	0.9444$_{\pm0.0521}$(4)	0.5277$_{\pm0.0682}$(12)
MO_GAAL	0.2743$_{\pm0.1774}$(24)	0.6858$_{\pm0.0735}$(**1**)	0.0050$_{\pm0.0063}$(25)	0.0074$_{\pm0.0074}$(24)	0.0320$_{\pm0.0388}$(24)	0.4373$_{\pm0.0912}$(24)
OCSVM	0.9027$_{\pm0.0260}$(16)	0.4294$_{\pm0.0843}$(19)	0.9960$_{\pm0.0045}$(4)	0.9887$_{\pm0.0057}$(10)	0.8410$_{\pm0.0944}$(18)	0.5137$_{\pm0.0344}$(15)
PCA	0.9197$_{\pm0.0195}$(14)	0.3973$_{\pm0.0598}$(24)	0.9907$_{\pm0.0091}$(10)	0.9868$_{\pm0.0058}$(14)	0.8450$_{\pm0.0888}$(17)	0.5096$_{\pm0.0543}$(17)
QMCD	0.9025$_{\pm0.0571}$(17)	0.4114$_{\pm0.0871}$(21)	0.3578$_{\pm0.1567}$(23)	0.5860$_{\pm0.1903}$(22)	0.6633$_{\pm0.1655}$(21)	0.5343$_{\pm0.0571}$(8)
SO_GAAL	0.2334$_{\pm0.1414}$(25)	0.6801$_{\pm0.0693}$(2)	0.0056$_{\pm0.0065}$(24)	0.0060$_{\pm0.0051}$(25)	0.0248$_{\pm0.0276}$(25)	0.4272$_{\pm0.0797}$(25)
Sampling	0.9210$_{\pm0.0424}$(13)	0.4449$_{\pm0.1259}$(16)	0.9871$_{\pm0.0271}$(13)	0.9817$_{\pm0.0095}$(17)	0.9328$_{\pm0.0487}$(9)	0.5505$_{\pm0.0742}$(5)
TCCM	0.8880$_{\pm0.0393}$(18)	0.5183$_{\pm0.0730}$(9)	0.9960$_{\pm0.0063}$(5)	0.9899$_{\pm0.0077}$(7)	0.9261$_{\pm0.0498}$(12)	0.4841$_{\pm0.0688}$(22)
VAE	0.9404$_{\pm0.0175}$(8)	0.5244$_{\pm0.0725}$(7)	0.9873$_{\pm0.0166}$(12)	0.9897$_{\pm0.0076}$(8)	0.9259$_{\pm0.0335}$(13)	0.5083$_{\pm0.0451}$(18)

TABLE XIII: ROC–AUC, medium datasets (1/3).

Model	2_annthyroid	6_cardio	7_Cardiotocography	12_fault	19_landsat	20_letter
ABOD	0.7439$_{\pm0.0124}$(15)	0.5877$_{\pm0.0275}$(21)	0.4917$_{\pm0.0184}$(23)	0.6928$_{\pm0.0110}$(12)	0.5164$_{\pm0.0127}$(19)	0.8652$_{\pm0.0149}$(4)
AutoEncoder	0.8612$_{\pm0.0192}$(6)	0.9307$_{\pm0.0271}$(14)	0.7202$_{\pm0.0335}$(14)	0.7386$_{\pm0.0188}$(6)	0.5773$_{\pm0.0111}$(15)	0.8135$_{\pm0.0340}$(9)
CBLOF	0.6578$_{\pm0.0179}$(19)	0.9546$_{\pm0.0043}$(7)	0.7815$_{\pm0.0315}$(8)	0.7094$_{\pm0.0358}$(10)	0.6907$_{\pm0.0179}$(4)	0.7755$_{\pm0.0370}$(11)
DIF	0.8025$_{\pm0.0282}$(12)	0.9570$_{\pm0.0075}$(6)	0.6877$_{\pm0.0171}$(17)	0.7386$_{\pm0.0145}$(7)	0.5947$_{\pm0.0167}$(13)	0.6661$_{\pm0.0575}$(14)
DiffInt-DA	0.8519$_{\pm0.0396}$(7)	0.9440$_{\pm0.0094}$(12)	0.7536$_{\pm0.0109}$(10)	0.7665$_{\pm0.0081}$(4)	0.6779$_{\pm0.0130}$(6)	0.8367$_{\pm0.0116}$(7)
DiffInt-IA	0.8251$_{\pm0.0222}$(10)	0.9423$_{\pm0.0074}$(13)	0.7498$_{\pm0.0128}$(12)	0.7789$_{\pm0.0089}$(3)	0.7170$_{\pm0.0090}$(2)	0.9231$_{\pm0.0135}$(2)
DiffInt-PA	0.8330$_{\pm0.0094}$(9)	0.9721$_{\pm0.0050}$(2)	0.8023$_{\pm0.0122}$(6)	0.7976$_{\pm0.0044}$(2)	0.6540$_{\pm0.0045}$(7)	0.9006$_{\pm0.0102}$(3)
ECOD	0.8249$_{\pm0.0099}$(11)	0.9519$_{\pm0.0047}$(10)	0.8322$_{\pm0.0096}$(3)	0.4919$_{\pm0.0126}$(22)	0.3959$_{\pm0.0111}$(23)	0.5871$_{\pm0.0418}$(19)
GMM	0.8412$_{\pm0.0270}$(8)	0.9448$_{\pm0.0075}$(11)	0.7507$_{\pm0.0134}$(11)	0.7024$_{\pm0.0160}$(11)	0.4967$_{\pm0.0096}$(21)	0.8473$_{\pm0.0350}$(6)
HBOS	0.7367$_{\pm0.0502}$(16)	0.8513$_{\pm0.0151}$(17)	0.6938$_{\pm0.0249}$(16)	0.6047$_{\pm0.0649}$(17)	0.6991$_{\pm0.0124}$(3)	0.5934$_{\pm0.0266}$(18)
IForest	0.9183$_{\pm0.0069}$(2)	0.9524$_{\pm0.0094}$(8)	0.8110$_{\pm0.0260}$(5)	0.6565$_{\pm0.0253}$(13)	0.6003$_{\pm0.0210}$(12)	0.6365$_{\pm0.0459}$(15)
KDE	0.5866$_{\pm0.0150}$(23)	0.9624$_{\pm0.0015}$(5)	0.8378$_{\pm0.0079}$(2)	0.7583$_{\pm0.0132}$(5)	0.5658$_{\pm0.0105}$(17)	0.7064$_{\pm0.0385}$(13)
KPCA	0.7473$_{\pm0.0161}$(14)	0.4696$_{\pm0.0601}$(23)	0.4918$_{\pm0.0441}$(22)	0.5139$_{\pm0.0176}$(21)	0.5017$_{\pm0.0162}$(20)	0.5057$_{\pm0.0391}$(21)
LMDD	0.5926$_{\pm0.0166}$(22)	0.5774$_{\pm0.0236}$(22)	0.7130$_{\pm0.0214}$(15)	0.5351$_{\pm0.0282}$(20)	0.4150$_{\pm0.0511}$(22)	0.4388$_{\pm0.0308}$(23)
LOF	0.7080$_{\pm0.0130}$(17)	0.7194$_{\pm0.0416}$(19)	0.5736$_{\pm0.0207}$(20)	0.6000$_{\pm0.0126}$(18)	0.5340$_{\pm0.0131}$(18)	0.8203$_{\pm0.0272}$(8)
LUNAR	0.8971$_{\pm0.0168}$(3)	0.9631$_{\pm0.0038}$(4)	0.7980$_{\pm0.0203}$(7)	0.8110$_{\pm0.0120}$(**1**)	0.7870$_{\pm0.0066}$(**1**)	0.9319$_{\pm0.0166}$(**1**)
MCD	0.9216$_{\pm0.0065}$(**1**)	0.8603$_{\pm0.0172}$(16)	0.6641$_{\pm0.0230}$(18)	0.6222$_{\pm0.0350}$(15)	0.6155$_{\pm0.0053}$(10)	0.8111$_{\pm0.0230}$(10)
MO_GAAL	0.5521$_{\pm0.0633}$(24)	0.2338$_{\pm0.1069}$(25)	0.4305$_{\pm0.1216}$(24)	0.4666$_{\pm0.0522}$(24)	0.6447$_{\pm0.0239}$(9)	0.4041$_{\pm0.0506}$(24)
OCSVM	0.5951$_{\pm0.0141}$(21)	0.9521$_{\pm0.0034}$(9)	0.8493$_{\pm0.0079}$(**1**)	0.5551$_{\pm0.0202}$(19)	0.3788$_{\pm0.0105}$(24)	0.5003$_{\pm0.0482}$(22)
PCA	0.6734$_{\pm0.0175}$(18)	0.7263$_{\pm0.2385}$(18)	0.5786$_{\pm0.1266}$(19)	0.4850$_{\pm0.0191}$(23)	0.3652$_{\pm0.0112}$(25)	0.5287$_{\pm0.0401}$(20)
QMCD	0.7681$_{\pm0.0130}$(13)	0.7117$_{\pm0.0402}$(20)	0.5658$_{\pm0.0256}$(21)	0.6437$_{\pm0.0409}$(14)	0.6834$_{\pm0.0094}$(5)	0.6183$_{\pm0.0358}$(17)
SO_GAAL	0.5461$_{\pm0.0616}$(25)	0.2399$_{\pm0.1106}$(24)	0.4256$_{\pm0.1214}$(25)	0.4652$_{\pm0.0450}$(25)	0.5759$_{\pm0.0248}$(16)	0.4014$_{\pm0.0512}$(25)
Sampling	0.6445$_{\pm0.0226}$(20)	0.8934$_{\pm0.0399}$(15)	0.7324$_{\pm0.0492}$(13)	0.7165$_{\pm0.0336}$(9)	0.6522$_{\pm0.0704}$(8)	0.7229$_{\pm0.0388}$(12)
TCCM	0.8840$_{\pm0.1138}$(4)	0.9775$_{\pm0.0094}$(**1**)	0.7653$_{\pm0.0441}$(9)	0.7362$_{\pm0.0257}$(8)	0.6044$_{\pm0.0186}$(11)	0.8574$_{\pm0.0287}$(5)
VAE	0.8821$_{\pm0.0227}$(5)	0.9673$_{\pm0.0092}$(3)	0.8153$_{\pm0.0201}$(4)	0.6082$_{\pm0.0315}$(16)	0.5807$_{\pm0.0081}$(14)	0.6259$_{\pm0.0420}$(16)

TABLE XIV: ROC–AUC, medium datasets (2/3).

Model	27_PageBlocks	28_pendigits	30_satellite	31_satimage-2	38_thyroid	40_vowels
ABOD	$0.7166_{\pm 0.0251}$(22)	$0.6559_{\pm 0.0303}$(20)	$0.5619_{\pm 0.0148}$(22)	$0.7878_{\pm 0.0471}$(20)	$0.9111_{\pm 0.0147}$(18)	$0.9440_{\pm 0.0194}$(8)
AutoEncoder	$0.9434_{\pm 0.0122}$(4)	$0.9805_{\pm 0.0078}$(7)	$0.8047_{\pm 0.0102}$(10)	$0.9975_{\pm 0.0020}$(9)	$0.9816_{\pm 0.0080}$(7)	$0.9460_{\pm 0.0187}$(7)
CBLOF	$0.8792_{\pm 0.0185}$(16)	$0.9760_{\pm 0.0054}$(9)	$0.8575_{\pm 0.0259}$(4)	$0.9985_{\pm 0.0019}$(5)	$0.9354_{\pm 0.0105}$(16)	$0.9343_{\pm 0.0234}$(9)
DIF	$0.9354_{\pm 0.0115}$(5)	$0.9815_{\pm 0.0070}$(6)	$0.7886_{\pm 0.0133}$(14)	$0.9979_{\pm 0.0017}$(6)	$0.9838_{\pm 0.0039}$(6)	$0.8136_{\pm 0.0444}$(12)
DiffInt-DA	$0.9314_{\pm 0.0094}$(7)	$0.9964_{\pm 0.0018}$(3)	$0.8472_{\pm 0.0048}$(5)	$0.9998_{\pm 0.0002}$(**1**)	$0.9571_{\pm 0.0138}$(11)	$0.9483_{\pm 0.0058}$(5)
DiffInt-IA	$0.9312_{\pm 0.0089}$(8)	$0.9948_{\pm 0.0036}$(4)	$0.8450_{\pm 0.0083}$(6)	$0.9989_{\pm 0.0018}$(3)	$0.9512_{\pm 0.0124}$(13)	$0.9665_{\pm 0.0064}$(3)
DiffInt-PA	$0.9500_{\pm 0.0043}$(2)	$0.9966_{\pm 0.0011}$(2)	$0.8416_{\pm 0.0046}$(7)	$0.9993_{\pm 0.0010}$(2)	$0.9770_{\pm 0.0047}$(8)	$0.9808_{\pm 0.0068}$(2)
ECOD	$0.9253_{\pm 0.0030}$(9)	$0.9329_{\pm 0.0094}$(16)	$0.6155_{\pm 0.0115}$(18)	$0.9749_{\pm 0.0169}$(19)	$0.9859_{\pm 0.0025}$(4)	$0.6123_{\pm 0.0509}$(20)
GMM	$0.9577_{\pm 0.0054}$(**1**)	$0.8485_{\pm 0.0120}$(18)	$0.8028_{\pm 0.0065}$(12)	$0.9958_{\pm 0.0013}$(12)	$0.9760_{\pm 0.0050}$(10)	$0.9480_{\pm 0.0177}$(6)
HBOS	$0.8133_{\pm 0.0120}$(18)	$0.9377_{\pm 0.0095}$(14)	$0.8624_{\pm 0.0077}$(2)	$0.9841_{\pm 0.0135}$(17)	$0.9877_{\pm 0.0039}$(3)	$0.7046_{\pm 0.0568}$(16)
IForest	$0.9200_{\pm 0.0099}$(10)	$0.9695_{\pm 0.0051}$(10)	$0.7967_{\pm 0.0151}$(13)	$0.9951_{\pm 0.0048}$(13)	$0.9878_{\pm 0.0034}$(2)	$0.7874_{\pm 0.0307}$(13)
KDE	$0.8868_{\pm 0.0099}$(15)	$0.9918_{\pm 0.0025}$(5)	$0.7678_{\pm 0.0099}$(15)	$0.9977_{\pm 0.0031}$(8)	$0.8526_{\pm 0.0155}$(20)	$0.7116_{\pm 0.0410}$(15)
KPCA	$0.6020_{\pm 0.0179}$(23)	$0.5025_{\pm 0.0867}$(22)	$0.4778_{\pm 0.0156}$(23)	$0.4588_{\pm 0.1931}$(23)	$0.3309_{\pm 0.0614}$(25)	$0.4887_{\pm 0.0458}$(23)
LMDD	$0.7235_{\pm 0.0263}$(21)	$0.6073_{\pm 0.0723}$(21)	$0.4468_{\pm 0.0678}$(24)	$0.5015_{\pm 0.1118}$(22)	$0.7074_{\pm 0.0709}$(22)	$0.5107_{\pm 0.0551}$(22)
LOF	$0.7654_{\pm 0.0132}$(19)	$0.4669_{\pm 0.0438}$(23)	$0.5634_{\pm 0.0141}$(21)	$0.4551_{\pm 0.2150}$(24)	$0.9417_{\pm 0.0230}$(15)	$0.9497_{\pm 0.0148}$(4)
LUNAR	$0.9353_{\pm 0.0114}$(6)	$0.9994_{\pm 0.0005}$(**1**)	$0.8803_{\pm 0.0073}$(**1**)	$0.9978_{\pm 0.0024}$(7)	$0.9770_{\pm 0.0053}$(9)	$0.9887_{\pm 0.0052}$(**1**)
MCD	$0.9195_{\pm 0.0075}$(11)	$0.8298_{\pm 0.0103}$(19)	$0.8093_{\pm 0.0073}$(9)	$0.9960_{\pm 0.0010}$(11)	$0.9852_{\pm 0.0029}$(5)	$0.7523_{\pm 0.0651}$(14)
MO_GAAL	$0.2884_{\pm 0.0932}$(25)	$0.2023_{\pm 0.0833}$(24)	$0.4177_{\pm 0.0347}$(25)	$0.2075_{\pm 0.1385}$(25)	$0.7068_{\pm 0.0820}$(23)	$0.1081_{\pm 0.0740}$(24)
OCSVM	$0.8962_{\pm 0.0092}$(14)	$0.9643_{\pm 0.0051}$(11)	$0.6537_{\pm 0.0126}$(17)	$0.9859_{\pm 0.0140}$(16)	$0.8610_{\pm 0.0151}$(19)	$0.5997_{\pm 0.0462}$(21)
PCA	$0.9061_{\pm 0.0075}$(13)	$0.9360_{\pm 0.0066}$(15)	$0.5999_{\pm 0.0117}$(20)	$0.9820_{\pm 0.0155}$(18)	$0.9571_{\pm 0.0053}$(12)	$0.6158_{\pm 0.0514}$(19)
QMCD	$0.7648_{\pm 0.0253}$(20)	$0.9106_{\pm 0.0099}$(17)	$0.8605_{\pm 0.0072}$(3)	$0.9961_{\pm 0.0036}$(10)	$0.8329_{\pm 0.0760}$(21)	$0.7020_{\pm 0.0460}$(17)
SO_GAAL	$0.2970_{\pm 0.0840}$(24)	$0.1966_{\pm 0.0956}$(25)	$0.6032_{\pm 0.1301}$(19)	$0.7099_{\pm 0.2435}$(21)	$0.6946_{\pm 0.0803}$(24)	$0.0992_{\pm 0.0719}$(25)
Sampling	$0.8629_{\pm 0.0352}$(17)	$0.9379_{\pm 0.0366}$(13)	$0.8141_{\pm 0.0294}$(8)	$0.9946_{\pm 0.0072}$(14)	$0.9239_{\pm 0.0183}$(17)	$0.8896_{\pm 0.0486}$(11)
TCCM	$0.9152_{\pm 0.0262}$(12)	$0.9774_{\pm 0.0198}$(8)	$0.8029_{\pm 0.0111}$(11)	$0.9989_{\pm 0.0014}$(4)	$0.9443_{\pm 0.0065}$(14)	$0.9339_{\pm 0.0172}$(10)
VAE	$0.9477_{\pm 0.0099}$(3)	$0.9454_{\pm 0.0062}$(12)	$0.7602_{\pm 0.0093}$(16)	$0.9872_{\pm 0.0081}$(15)	$0.9883_{\pm 0.0027}$(**1**)	$0.6742_{\pm 0.0538}$(18)

TABLE XV: ROC–AUC, medium datasets (3/3).

Model	41_Waveform	44_Wilt	47_yeast
ABOD	$0.6265_{\pm 0.0327}$(16)	$0.5544_{\pm 0.0226}$(11)	$0.4188_{\pm 0.0212}$(20)
AutoEncoder	$0.6848_{\pm 0.0565}$(13)	$0.5556_{\pm 0.0760}$(10)	$0.4561_{\pm 0.0156}$(11)
CBLOF	$0.7546_{\pm 0.0522}$(3)	$0.3755_{\pm 0.0282}$(21)	$0.4777_{\pm 0.0209}$(6)
DIF	$0.7470_{\pm 0.0248}$(5)	$0.4179_{\pm 0.0614}$(18)	$0.3969_{\pm 0.0183}$(25)
DiffInt-DA	$0.6076_{\pm 0.0143}$(18)	$0.8616_{\pm 0.0774}$(2)	$0.4719_{\pm 0.0175}$(7)
DiffInt-IA	$0.6884_{\pm 0.0250}$(12)	$0.9033_{\pm 0.0180}$(**1**)	$0.4583_{\pm 0.0130}$(10)
DiffInt-PA	$0.7678_{\pm 0.0178}$(**1**)	$0.6789_{\pm 0.0272}$(7)	$0.4926_{\pm 0.0163}$(5)
ECOD	$0.6140_{\pm 0.0352}$(17)	$0.3994_{\pm 0.0286}$(20)	$0.4492_{\pm 0.0167}$(14)
GMM	$0.5790_{\pm 0.0356}$(19)	$0.7295_{\pm 0.0295}$(4)	$0.4374_{\pm 0.0190}$(16)
HBOS	$0.6973_{\pm 0.0316}$(11)	$0.4051_{\pm 0.0995}$(19)	$0.4213_{\pm 0.0180}$(19)
IForest	$0.7332_{\pm 0.0412}$(8)	$0.4687_{\pm 0.0343}$(12)	$0.4106_{\pm 0.0163}$(21)
KDE	$0.7661_{\pm 0.0269}$(2)	$0.3569_{\pm 0.0299}$(23)	$0.4034_{\pm 0.0169}$(22)
KPCA	$0.5138_{\pm 0.0243}$(23)	$0.6498_{\pm 0.0243}$(8)	$0.5649_{\pm 0.0142}$(**1**)
LMDD	$0.5173_{\pm 0.0232}$(22)	$0.4192_{\pm 0.0570}$(17)	$0.4642_{\pm 0.0230}$(8)
LOF	$0.7395_{\pm 0.0396}$(7)	$0.4265_{\pm 0.0151}$(15)	$0.4327_{\pm 0.0215}$(17)
LUNAR	$0.7528_{\pm 0.0484}$(4)	$0.5917_{\pm 0.0676}$(9)	$0.4477_{\pm 0.0207}$(15)
MCD	$0.5783_{\pm 0.0365}$(20)	$0.8608_{\pm 0.0102}$(3)	$0.4541_{\pm 0.0228}$(12)
MO_GAAL	$0.3040_{\pm 0.0835}$(25)	$0.6966_{\pm 0.0301}$(5)	$0.5599_{\pm 0.0385}$(3)
OCSVM	$0.5662_{\pm 0.0422}$(21)	$0.3574_{\pm 0.0313}$(22)	$0.4260_{\pm 0.0158}$(18)
PCA	$0.6432_{\pm 0.0373}$(14)	$0.4404_{\pm 0.0234}$(14)	$0.3982_{\pm 0.0167}$(24)
QMCD	$0.7448_{\pm 0.0245}$(6)	$0.3565_{\pm 0.0351}$(25)	$0.3991_{\pm 0.0196}$(23)
SO_GAAL	$0.3466_{\pm 0.0815}$(24)	$0.6906_{\pm 0.0275}$(6)	$0.5642_{\pm 0.0432}$(2)
Sampling	$0.7022_{\pm 0.0578}$(10)	$0.4217_{\pm 0.0380}$(16)	$0.4602_{\pm 0.0513}$(9)
TCCM	$0.6406_{\pm 0.0827}$(15)	$0.3567_{\pm 0.0623}$(24)	$0.5201_{\pm 0.0587}$(4)
VAE	$0.7081_{\pm 0.0266}$(9)	$0.4547_{\pm 0.0278}$(13)	$0.4500_{\pm 0.0162}$(13)

TABLE XVI: ROC–AUC, large datasets (1/2).

Model	1_ALOI	8_celeba	10_cover	11_donors	13_fraud	16_http
ABOD	$0.7311_{\pm 0.0082}$(3)	$0.4706_{\pm 0.0053}$(21)	$0.7391_{\pm 0.0072}$(18)	$0.4036_{\pm 0.0014}$(18)	$0.8889_{\pm 0.0172}$(19)	$0.7510_{\pm 0.0017}$(18)
AutoEncoder	$0.5554_{\pm 0.0126}$(6)	$0.6988_{\pm 0.0311}$(12)	$0.9843_{\pm 0.0044}$(4)	$0.9359_{\pm 0.0100}$(5)	$0.9592_{\pm 0.0070}$(4)	$0.9988_{\pm 0.0012}$(8)
CBLOF	$0.5449_{\pm 0.0101}$(13)	$0.6790_{\pm 0.0629}$(13)	$0.9245_{\pm 0.0174}$(11)	$0.9159_{\pm 0.0156}$(8)	$0.9535_{\pm 0.0052}$(9)	$0.9949_{\pm 0.0001}$(14)
DIF	$0.5498_{\pm 0.0103}$(10)	$0.6636_{\pm 0.0155}$(15)	$0.9787_{\pm 0.0040}$(5)	$0.9118_{\pm 0.0101}$(10)	$0.9528_{\pm 0.0078}$(10)	$0.9935_{\pm 0.0002}$(16)
DiffInt-DA	$0.5552_{\pm 0.0092}$(7)	$0.6535_{\pm 0.0540}$(16)	$0.9899_{\pm 0.0058}$(2)	$1.0000_{\pm 0.0000}$(1)	$0.9459_{\pm 0.0047}$(16)	$0.9998_{\pm 0.0006}$(2)
DiffInt-IA	$0.5483_{\pm 0.0070}$(12)	$0.7200_{\pm 0.0253}$(9)	$0.9866_{\pm 0.0051}$(3)	$0.9999_{\pm 0.0000}$(3)	$0.9505_{\pm 0.0055}$(13)	$0.9997_{\pm 0.0008}$(4)
DiffInt-PA	$0.5857_{\pm 0.0067}$(4)	$0.7463_{\pm 0.0386}$(8)	$0.9949_{\pm 0.0015}$(**1**)	$1.0000_{\pm 0.0000}$(**1**)	$0.9596_{\pm 0.0025}$(3)	$0.9998_{\pm 0.0022}$(**1**)
ECOD	$0.5326_{\pm 0.0086}$(18)	$0.7664_{\pm 0.0030}$(7)	$0.9386_{\pm 0.0019}$(8)	$0.9215_{\pm 0.0005}$(7)	$0.9525_{\pm 0.0057}$(11)	$0.9880_{\pm 0.0003}$(17)
GMM	$0.5601_{\pm 0.0100}$(5)	$0.8076_{\pm 0.0032}$(3)	$0.9505_{\pm 0.0012}$(7)	$0.9255_{\pm 0.0004}$(6)	$0.9597_{\pm 0.0053}$(2)	$0.9994_{\pm 0.0000}$(6)
HBOS	$0.5323_{\pm 0.0091}$(19)	$0.7685_{\pm 0.0031}$(6)	$0.7880_{\pm 0.0041}$(17)	$0.7922_{\pm 0.0016}$(15)	$0.9551_{\pm 0.0060}$(7)	$0.9961_{\pm 0.0014}$(10)
IForest	$0.5382_{\pm 0.0112}$(14)	$0.7127_{\pm 0.0194}$(10)	$0.8456_{\pm 0.0235}$(14)	$0.9124_{\pm 0.0222}$(9)	$0.9513_{\pm 0.0063}$(12)	$0.9938_{\pm 0.0029}$(15)
KDE	$0.5367_{\pm 0.0103}$(16)	$0.6675_{\pm 0.0039}$(14)	$0.8716_{\pm 0.0019}$(12)	-	-	-
KPCA	$0.5327_{\pm 0.0114}$(17)	-	-	-	-	-
LMDD	$0.5322_{\pm 0.0124}$(20)	-	-	-	-	-
LOF	$0.7347_{\pm 0.0109}$(2)	$0.4171_{\pm 0.0091}$(22)	$0.5547_{\pm 0.0107}$(20)	$0.6111_{\pm 0.0036}$(17)	$0.4976_{\pm 0.0086}$(20)	$0.3929_{\pm 0.0052}$(20)
LUNAR	$0.7389_{\pm 0.0114}$(**1**)	$0.6335_{\pm 0.0069}$(17)	-	-	$0.9658_{\pm 0.0057}$(**1**)	$0.9953_{\pm 0.0037}$(12)
MCD	$0.5262_{\pm 0.0088}$(22)	$0.8246_{\pm 0.0175}$(2)	$0.7027_{\pm 0.0025}$(19)	$0.8202_{\pm 0.0993}$(14)	$0.9232_{\pm 0.0077}$(18)	$0.9995_{\pm 0.0000}$(5)
MO_GAAL	$0.4740_{\pm 0.0121}$(25)	$0.2365_{\pm 0.0809}$(23)	-	-	-	-
OCSVM	$0.5320_{\pm 0.0086}$(21)	$0.7100_{\pm 0.0033}$(11)	$0.9251_{\pm 0.0020}$(10)	-	$0.9501_{\pm 0.0072}$(14)	-
PCA	$0.5485_{\pm 0.0107}$(11)	$0.7851_{\pm 0.0032}$(4)	$0.9337_{\pm 0.0010}$(9)	$0.8240_{\pm 0.0011}$(13)	$0.9546_{\pm 0.0052}$(8)	$0.9961_{\pm 0.0001}$(11)
QMCD	$0.5258_{\pm 0.0096}$(23)	$0.5000_{\pm 0.0000}$(19)	$0.8215_{\pm 0.0030}$(16)	$0.7250_{\pm 0.0033}$(16)	$0.9575_{\pm 0.0059}$(5)	$0.9968_{\pm 0.0002}$(9)
SO_GAAL	$0.4770_{\pm 0.0113}$(24)	$0.4803_{\pm 0.0248}$(20)	$0.5161_{\pm 0.0088}$(21)	$0.3832_{\pm 0.1452}$(19)	$0.0751_{\pm 0.0216}$(21)	$0.4447_{\pm 0.1537}$(19)
Sampling	$0.5376_{\pm 0.0108}$(15)	$0.6126_{\pm 0.1041}$(18)	$0.8391_{\pm 0.0579}$(15)	$0.8793_{\pm 0.0257}$(11)	$0.9472_{\pm 0.0106}$(15)	$0.9952_{\pm 0.0014}$(13)
TCCM	$0.5501_{\pm 0.0163}$(9)	$0.8308_{\pm 0.0109}$(**1**)	$0.8705_{\pm 0.1019}$(13)	$0.9955_{\pm 0.0057}$(4)	$0.9404_{\pm 0.0136}$(17)	$0.9997_{\pm 0.0007}$(3)
VAE	$0.5541_{\pm 0.0084}$(8)	$0.7690_{\pm 0.0035}$(5)	$0.9745_{\pm 0.0009}$(6)	$0.8428_{\pm 0.0458}$(12)	$0.9574_{\pm 0.0045}$(6)	$0.9994_{\pm 0.0000}$(7)

TABLE XVII: ROC–AUC, large datasets (2/2).

Model	22_magic	23_mammography	32_shuttle	33_skin	34_smtp
ABOD	$0.7930_{\pm 0.0069}$(9)	$0.5424_{\pm 0.0121}$(23)	$0.6135_{\pm 0.0077}$(22)	$0.4064_{\pm 0.0026}$(22)	$0.8606_{\pm 0.0198}$(10)
AutoEncoder	$0.8211_{\pm 0.0111}$(6)	$0.9114_{\pm 0.0197}$(2)	$0.9968_{\pm 0.0010}$(5)	$0.7975_{\pm 0.0685}$(11)	$0.9285_{\pm 0.0433}$(2)
CBLOF	$0.8235_{\pm 0.0062}$(5)	$0.8238_{\pm 0.0188}$(15)	$0.9871_{\pm 0.0029}$(17)	$0.9178_{\pm 0.0035}$(5)	$0.8446_{\pm 0.0747}$(16)
DIF	$0.7822_{\pm 0.0136}$(10)	$0.8281_{\pm 0.0231}$(14)	$0.9905_{\pm 0.0023}$(11)	$0.8759_{\pm 0.0087}$(10)	$0.8490_{\pm 0.0617}$(15)
DiffInt-DA	$0.8334_{\pm 0.0104}$(3)	$0.8464_{\pm 0.0231}$(13)	$0.9984_{\pm 0.0003}$(4)	$0.9345_{\pm 0.0247}$(3)	$0.8792_{\pm 0.0342}$(8)
DiffInt-IA	$0.8288_{\pm 0.0072}$(4)	$0.8544_{\pm 0.0128}$(10)	$0.9987_{\pm 0.0003}$(3)	$0.9311_{\pm 0.0164}$(4)	$0.8565_{\pm 0.0496}$(13)
DiffInt-PA	$0.8644_{\pm 0.0026}$(**1**)	$0.8803_{\pm 0.0097}$(7)	$0.9992_{\pm 0.0004}$(2)	$0.9909_{\pm 0.0021}$(**1**)	$0.8961_{\pm 0.0151}$(6)
ECOD	$0.6541_{\pm 0.0055}$(21)	$0.9115_{\pm 0.0163}$(**1**)	$0.9962_{\pm 0.0007}$(7)	$0.5421_{\pm 0.0011}$(20)	$0.8868_{\pm 0.0419}$(7)
GMM	$0.8040_{\pm 0.0036}$(7)	$0.8814_{\pm 0.0198}$(6)	$0.9936_{\pm 0.0016}$(10)	$0.8885_{\pm 0.0004}$(8)	$0.8311_{\pm 0.0862}$(18)
HBOS	$0.7301_{\pm 0.0060}$(15)	$0.8706_{\pm 0.0169}$(8)	$0.9877_{\pm 0.0037}$(15)	$0.7722_{\pm 0.0011}$(12)	$0.8528_{\pm 0.0687}$(14)
IForest	$0.7764_{\pm 0.0148}$(11)	$0.8832_{\pm 0.0230}$(4)	$0.9965_{\pm 0.0008}$(6)	$0.8909_{\pm 0.0043}$(7)	$0.9061_{\pm 0.0377}$(5)
KDE	$0.7624_{\pm 0.0043}$(13)	$0.8581_{\pm 0.0153}$(9)	$0.9875_{\pm 0.0021}$(16)	$0.7324_{\pm 0.0012}$(14)	$0.8026_{\pm 0.0518}$(19)
KPCA	$0.5247_{\pm 0.0061}$(22)	$0.7860_{\pm 0.0135}$(19)	$0.9270_{\pm 0.0039}$(19)	-	-
LMDD	$0.5204_{\pm 0.0306}$(23)	$0.7440_{\pm 0.0772}$(20)	$0.6638_{\pm 0.1080}$(21)	-	$0.6184_{\pm 0.1197}$(22)
LOF	$0.6790_{\pm 0.0044}$(19)	$0.8018_{\pm 0.0235}$(17)	$0.5391_{\pm 0.0107}$(23)	$0.5472_{\pm 0.0032}$(19)	$0.9139_{\pm 0.0409}$(4)
LUNAR	$0.8554_{\pm 0.0047}$(2)	$0.8817_{\pm 0.0209}$(5)	$0.9997_{\pm 0.0002}$(**1**)	$0.9901_{\pm 0.0015}$(2)	$0.9280_{\pm 0.0266}$(3)
MCD	$0.7380_{\pm 0.0043}$(14)	$0.7199_{\pm 0.1363}$(22)	$0.9898_{\pm 0.0009}$(13)	$0.8844_{\pm 0.0004}$(9)	$0.9508_{\pm 0.0149}$(**1**)
MO_GAAL	$0.2031_{\pm 0.0117}$(25)	$0.1150_{\pm 0.0270}$(24)	$0.0289_{\pm 0.0036}$(25)	-	$0.4082_{\pm 0.0658}$(24)
OCSVM	$0.7030_{\pm 0.0048}$(18)	$0.8486_{\pm 0.0127}$(12)	$0.9869_{\pm 0.0022}$(18)	$0.7610_{\pm 0.0013}$(13)	$0.7982_{\pm 0.0538}$(20)
PCA	$0.6666_{\pm 0.0062}$(20)	$0.8923_{\pm 0.0151}$(3)	$0.9899_{\pm 0.0017}$(12)	$0.4149_{\pm 0.0026}$(21)	$0.8348_{\pm 0.0519}$(17)
QMCD	$0.7157_{\pm 0.0053}$(16)	$0.7313_{\pm 0.0276}$(21)	$0.7274_{\pm 0.0097}$(20)	$0.6219_{\pm 0.0010}$(17)	$0.7361_{\pm 0.0453}$(21)
SO_GAAL	$0.2044_{\pm 0.0145}$(24)	$0.1149_{\pm 0.0253}$(25)	$0.0297_{\pm 0.0039}$(24)	$0.5628_{\pm 0.2584}$(18)	$0.4270_{\pm 0.1116}$(23)
Sampling	$0.7934_{\pm 0.0237}$(8)	$0.8047_{\pm 0.0303}$(16)	$0.9892_{\pm 0.0030}$(14)	$0.9165_{\pm 0.0304}$(6)	$0.8701_{\pm 0.0589}$(9)
TCCM	$0.7748_{\pm 0.0216}$(12)	$0.8503_{\pm 0.0335}$(11)	$0.9961_{\pm 0.0007}$(8)	$0.6414_{\pm 0.2798}$(15)	$0.8591_{\pm 0.0747}$(12)
VAE	$0.7088_{\pm 0.0183}$(17)	$0.7924_{\pm 0.0272}$(18)	$0.9948_{\pm 0.0012}$(9)	$0.6267_{\pm 0.0018}$(16)	$0.8595_{\pm 0.0712}$(11)

TABLE XVIII: ROC–AUC, high-dimensional datasets (1/2).

Model	3_backdoor	5_campaign	9_census	17_InternetAds	24_mnist	25_musk
ABOD	0.6392$_{\pm0.0091}$(19)	0.6588$_{\pm0.0065}$(18)	0.4026$_{\pm0.0026}$(12)	0.6577$_{\pm0.0328}$(15)	0.7120$_{\pm0.0145}$(18)	0.2821$_{\pm0.0881}$(22)
AutoEncoder	0.9331$_{\pm0.0084}$(7)	0.8128$_{\pm0.0074}$(**1**)	-	0.8806$_{\pm0.0147}$(3)	0.9345$_{\pm0.0081}$(**1**)	1.0000$_{\pm0.0000}$(**1**)
CBLOF	0.8396$_{\pm0.0204}$(14)	0.6743$_{\pm0.0125}$(17)	-	0.7185$_{\pm0.0115}$(7)	0.8727$_{\pm0.0109}$(13)	1.0000$_{\pm0.0000}$(**1**)
DIF	0.9311$_{\pm0.0067}$(8)	0.6776$_{\pm0.0116}$(16)	-	0.6346$_{\pm0.0217}$(18)	0.8818$_{\pm0.0097}$(10)	0.9999$_{\pm0.0001}$(13)
DiffInt-DA	0.9682$_{\pm0.0083}$(**1**)	0.8064$_{\pm0.0066}$(3)	0.7082$_{\pm0.0036}$(3)	0.7115$_{\pm0.0060}$(9)	0.9301$_{\pm0.0055}$(4)	1.0000$_{\pm0.0000}$(**1**)
DiffInt-IA	0.9650$_{\pm0.0060}$(2)	0.8010$_{\pm0.0061}$(7)	0.6889$_{\pm0.0015}$(6)	0.6493$_{\pm0.0174}$(16)	0.9155$_{\pm0.0111}$(7)	1.0000$_{\pm0.0000}$(**1**)
DiffInt-PA	0.9613$_{\pm0.0043}$(3)	0.8106$_{\pm0.0050}$(2)	0.6273$_{\pm0.0007}$(7)	0.7036$_{\pm0.0053}$(12)	0.8950$_{\pm0.0074}$(8)	1.0000$_{\pm0.0000}$(**1**)
ECOD	0.8567$_{\pm0.0077}$(13)	0.7836$_{\pm0.0040}$(10)	-	0.7079$_{\pm0.0116}$(11)	0.7814$_{\pm0.0121}$(16)	0.9984$_{\pm0.0008}$(16)
GMM	0.9276$_{\pm0.0083}$(9)	0.7973$_{\pm0.0040}$(8)	-	0.9123$_{\pm0.0089}$(**1**)	0.9240$_{\pm0.0080}$(6)	1.0000$_{\pm0.0000}$(**1**)
HBOS	0.6620$_{\pm0.0077}$(18)	0.8011$_{\pm0.0046}$(6)	-	0.4937$_{\pm0.0488}$(23)	0.6706$_{\pm0.0110}$(19)	1.0000$_{\pm0.0000}$(**1**))
IForest	0.7612$_{\pm0.0352}$(17)	0.7285$_{\pm0.0159}$(12)	0.6185$_{\pm0.0120}$(8)	0.4246$_{\pm0.0516}$(25)	0.8645$_{\pm0.0181}$(14)	0.9409$_{\pm0.0387}$(18)
KDE	0.9386$_{\pm0.0068}$(6)	0.6999$_{\pm0.0065}$(14)	-	0.8524$_{\pm0.0118}$(6)	0.9336$_{\pm0.0067}$(2)	1.0000$_{\pm0.0000}$(**1**)
KPCA	-	0.4083$_{\pm0.0078}$(25)	-	0.5284$_{\pm0.0289}$(20)	0.5050$_{\pm0.0231}$(21)	0.5021$_{\pm0.0694}$(19)
LMDD	-	0.4491$_{\pm0.0581}$(22)	-	0.6700$_{\pm0.0126}$(14)	0.3897$_{\pm0.0343}$(25)	0.3448$_{\pm0.0445}$(21)
LOF	0.7669$_{\pm0.0131}$(16)	0.5922$_{\pm0.0063}$(20)	0.5308$_{\pm0.0024}$(9)	0.6397$_{\pm0.0372}$(17)	0.7276$_{\pm0.0241}$(17)	0.3703$_{\pm0.0383}$(20)
LUNAR	0.9539$_{\pm0.0064}$(4)	0.7266$_{\pm0.0085}$(13)	-	0.8835$_{\pm0.0133}$(2)	0.9286$_{\pm0.0082}$(5)	1.0000$_{\pm0.0000}$(**1**)
MCD	0.8659$_{\pm0.0879}$(12)	0.7961$_{\pm0.0082}$(9)	0.7426$_{\pm0.0106}$(**1**)	0.5165$_{\pm0.0600}$(21)	0.8790$_{\pm0.0245}$(11)	0.9998$_{\pm0.0005}$(14)
MO_GAAL	0.4091$_{\pm0.2470}$(22)	0.4278$_{\pm0.0608}$(24)	-	0.4862$_{\pm0.0435}$(24)	0.4932$_{\pm0.1136}$(24)	0.2547$_{\pm0.1279}$(24)
OCSVM	0.8668$_{\pm0.0070}$(11)	0.6903$_{\pm0.0052}$(15)	-	0.7079$_{\pm0.0117}$(10)	0.8840$_{\pm0.0109}$(9)	0.9927$_{\pm0.0040}$(17)
PCA	0.5000$_{\pm0.0000}$(21)	0.5000$_{\pm0.0000}$(21)	0.5000$_{\pm0.0000}$(10)	0.6157$_{\pm0.0151}$(19)	0.5000$_{\pm0.0000}$(23)	1.0000$_{\pm0.0000}$(**1**)
QMCD	0.1021$_{\pm0.0065}$(23)	0.8029$_{\pm0.0074}$(5)	0.0000$_{\pm0.0019}$(13)	0.8591$_{\pm0.0185}$(5)	0.6071$_{\pm0.0088}$(20)	0.0329$_{\pm0.0160}$(25)
SO_GAAL	0.5055$_{\pm0.1733}$(20)	0.4387$_{\pm0.0670}$(23)	0.4850$_{\pm0.0353}$(11)	0.5000$_{\pm0.0000}$(22)	0.5021$_{\pm0.1107}$(22)	0.2675$_{\pm0.1256}$(23)
Sampling	0.8221$_{\pm0.0878}$(15)	0.6508$_{\pm0.0233}$(19)	-	0.7121$_{\pm0.0163}$(8)	0.8177$_{\pm0.0664}$(15)	0.9997$_{\pm0.0004}$(15)
TCCM	0.9526$_{\pm0.0111}$(5)	0.8052$_{\pm0.0177}$(4)	0.6896$_{\pm0.0068}$(5)	0.6984$_{\pm0.0109}$(13)	0.8728$_{\pm0.0247}$(12)	1.0000$_{\pm0.0000}$(**1**)
VAE	0.9129$_{\pm0.0046}$(10)	0.7800$_{\pm0.0039}$(11)	0.7048$_{\pm0.0021}$(4)	0.8782$_{\pm0.0151}$(4)	0.9332$_{\pm0.0083}$(3)	1.0000$_{\pm0.0000}$(**1**)

TABLE XIX: ROC–AUC, high-dimensional datasets (2/2).

Model	26_optdigits	35_SpamBase	36_speech	arrhythmia
ABOD	0.4934$_{\pm0.0350}$(20)	0.4517$_{\pm0.0131}$(22)	0.6062$_{\pm0.0881}$(2)	0.7597$_{\pm0.0329}$(20)
AutoEncoder	0.8707$_{\pm0.0148}$(8)	0.8186$_{\pm0.0140}$(3)	0.4570$_{\pm0.0492}$(15)	0.7890$_{\pm0.0327}$(15)
CBLOF	0.8937$_{\pm0.0239}$(5)	0.6153$_{\pm0.0182}$(15)	0.4515$_{\pm0.0473}$(18)	0.8082$_{\pm0.0357}$(9)
DIF	0.6939$_{\pm0.0792}$(13)	0.5566$_{\pm0.0306}$(19)	0.4397$_{\pm0.0622}$(24)	0.8060$_{\pm0.0362}$(10)
DiffInt-DA	0.9569$_{\pm0.0105}$(4)	0.7014$_{\pm0.0111}$(13)	0.5123$_{\pm0.0264}$(4)	0.8446$_{\pm0.0163}$(2)
DiffInt-IA	0.7755$_{\pm0.0312}$(12)	0.7099$_{\pm0.0069}$(11)	0.5054$_{\pm0.0253}$(5)	0.8537$_{\pm0.0126}$(**1**)
DiffInt-PA	0.9604$_{\pm0.0072}$(3)	0.7041$_{\pm0.0072}$(12)	0.5047$_{\pm0.0308}$(6)	0.8193$_{\pm0.0185}$(7)
ECOD	0.6357$_{\pm0.0142}$(17)	0.7194$_{\pm0.0097}$(9)	0.4511$_{\pm0.0486}$(19)	0.8236$_{\pm0.0319}$(6)
GMM	0.8067$_{\pm0.0087}$(11)	0.7974$_{\pm0.0147}$(7)	0.5013$_{\pm0.0483}$(7)	0.7819$_{\pm0.0346}$(18)
HBOS	0.8763$_{\pm0.0137}$(6)	0.7924$_{\pm0.0236}$(8)	0.4550$_{\pm0.0472}$(16)	0.8287$_{\pm0.0309}$(3)
IForest	0.8421$_{\pm0.0192}$(9)	0.8394$_{\pm0.0139}$(**1**)	0.4673$_{\pm0.0339}$(12)	0.8247$_{\pm0.0345}$(4)
KDE	0.9969$_{\pm0.0009}$(**1**)	0.6064$_{\pm0.0140}$(17)	0.6512$_{\pm0.0442}$(**1**)	0.8150$_{\pm0.0298}$(8)
KPCA	0.4975$_{\pm0.0319}$(19)	0.5190$_{\pm0.0172}$(21)	0.4890$_{\pm0.0492}$(10)	0.4717$_{\pm0.0689}$(22)
LMDD	0.4243$_{\pm0.0279}$(22)	0.5985$_{\pm0.0418}$(18)	0.4750$_{\pm0.0355}$(11)	0.5193$_{\pm0.0305}$(21)
LOF	0.4926$_{\pm0.0308}$(21)	0.4242$_{\pm0.0150}$(23)	0.4666$_{\pm0.0494}$(13)	0.7974$_{\pm0.0365}$(12)
LUNAR	0.9966$_{\pm0.0015}$(2)	0.8229$_{\pm0.0234}$(2)	0.5499$_{\pm0.0944}$(3)	0.8239$_{\pm0.0310}$(5)
MCD	0.6460$_{\pm0.0212}$(15)	0.8001$_{\pm0.0260}$(6)	0.4956$_{\pm0.0460}$(8)	0.7812$_{\pm0.0400}$(19)
MO_GAAL	0.3687$_{\pm0.1581}$(23)	0.3253$_{\pm0.0590}$(24)	0.4483$_{\pm0.0608}$(21)	0.3027$_{\pm0.0591}$(24)
OCSVM	0.6456$_{\pm0.0172}$(16)	0.6178$_{\pm0.0142}$(14)	0.4459$_{\pm0.0481}$(22)	0.8005$_{\pm0.0279}$(11)
PCA	0.5000$_{\pm0.0000}$(18)	0.5500$_{\pm0.0125}$(20)	0.4510$_{\pm0.0486}$(20)	0.7858$_{\pm0.0339}$(17)
QMCD	0.1303$_{\pm0.0179}$(25)	0.7141$_{\pm0.0237}$(10)	0.4217$_{\pm0.0446}$(25)	0.3366$_{\pm0.0624}$(23)
SO_GAAL	0.3658$_{\pm0.1716}$(24)	0.3198$_{\pm0.0548}$(25)	0.4457$_{\pm0.0656}$(23)	0.3005$_{\pm0.0800}$(25)
Sampling	0.8181$_{\pm0.0723}$(10)	0.6115$_{\pm0.0168}$(16)	0.4644$_{\pm0.0582}$(14)	0.7964$_{\pm0.0368}$(13)
TCCM	0.8725$_{\pm0.1032}$(7)	0.8092$_{\pm0.0214}$(5)	0.4899$_{\pm0.0480}$(9)	0.7886$_{\pm0.0370}$(16)
VAE	0.6583$_{\pm0.0277}$(14)	0.8118$_{\pm0.0146}$(4)	0.4516$_{\pm0.0489}$(17)	0.7926$_{\pm0.0397}$(14)

TABLE XX: AUPRC, small datasets (1/2).

Model	4_breastw	14_glass	15_Hepatitis	18_Ionosphere	21_Lymphography	29_Pima
ABOD	$0.2877_{\pm 0.0169}$(22)	$0.2252_{\pm 0.1340}$(10)	$0.3422_{\pm 0.1897}$(21)	$0.9189_{\pm 0.0188}$(11)	$0.6225_{\pm 0.2143}$(20)	$0.4735_{\pm 0.0308}$(18)
AutoEncoder	$0.9664_{\pm 0.0184}$(15)	$0.1502_{\pm 0.1025}$(18)	$0.6348_{\pm 0.1385}$(2)	$0.9416_{\pm 0.0131}$(7)	$0.8167_{\pm 0.1820}$(17)	$0.5387_{\pm 0.0181}$(7)
CBLOF	$0.9892_{\pm 0.0043}$(4)	$0.2286_{\pm 0.1182}$(9)	$0.5388_{\pm 0.1740}$(5)	$0.9201_{\pm 0.0374}$(10)	$0.9299_{\pm 0.1435}$(12)	$0.5244_{\pm 0.0279}$(10)
DIF	$0.6048_{\pm 0.0525}$(19)	$0.2519_{\pm 0.1434}$(6)	$0.4093_{\pm 0.1430}$(19)	$0.9367_{\pm 0.0108}$(8)	$0.4838_{\pm 0.2684}$(21)	$0.4144_{\pm 0.0207}$(22)
DiffInt-DA	$0.9854_{\pm 0.0056}$(8)	$0.3523_{\pm 0.0807}$(2)	$0.5455_{\pm 0.0787}$(4)	$0.9436_{\pm 0.0120}$(6)	$0.9843_{\pm 0.0393}$(5)	$0.5061_{\pm 0.0144}$(11)
DiffInt-IA	$0.9803_{\pm 0.0059}$(10)	$0.4035_{\pm 0.0867}$(**1**)	$0.4982_{\pm 0.0581}$(11)	$0.9438_{\pm 0.0132}$(5)	$0.9423_{\pm 0.0530}$(10)	$0.4999_{\pm 0.0119}$(14)
DiffInt-PA	$0.9919_{\pm 0.0032}$(**1**)	$0.3294_{\pm 0.0800}$(3)	$0.5211_{\pm 0.0667}$(9)	$0.9574_{\pm 0.0073}$(2)	$0.9951_{\pm 0.0201}$(2)	$0.5668_{\pm 0.0149}$(3)
ECOD	$0.9869_{\pm 0.0053}$(7)	$0.1812_{\pm 0.1485}$(13)	$0.4933_{\pm 0.0920}$(12)	$0.6874_{\pm 0.0444}$(19)	$0.9867_{\pm 0.0292}$(4)	$0.4992_{\pm 0.0179}$(15)
GMM	$0.9693_{\pm 0.0139}$(14)	$0.2004_{\pm 0.1567}$(12)	$0.5098_{\pm 0.1610}$(10)	$0.9569_{\pm 0.0099}$(3)	$0.7933_{\pm 0.1614}$(18)	$0.5418_{\pm 0.0200}$(6)
HBOS	$0.9799_{\pm 0.0088}$(11)	$0.1794_{\pm 0.1047}$(14)	$0.5721_{\pm 0.0927}$(3)	$0.5333_{\pm 0.0867}$(21)	$0.9637_{\pm 0.0659}$(9)	$0.5991_{\pm 0.0253}$(2)
IForest	$0.9890_{\pm 0.0053}$(5)	$0.1543_{\pm 0.0638}$(16)	$0.5316_{\pm 0.1013}$(6)	$0.8529_{\pm 0.0338}$(15)	$0.9900_{\pm 0.0211}$(3)	$0.5572_{\pm 0.0257}$(4)
KDE	$0.9904_{\pm 0.0041}$(2)	$0.1668_{\pm 0.1048}$(15)	$0.4333_{\pm 0.0932}$(17)	$0.9323_{\pm 0.0105}$(9)	$0.9833_{\pm 0.0351}$(7)	$0.5295_{\pm 0.0295}$(9)
KPCA	$0.2457_{\pm 0.0273}$(23)	$0.0525_{\pm 0.0186}$(25)	$0.2266_{\pm 0.1709}$(23)	$0.3667_{\pm 0.0443}$(23)	$0.1209_{\pm 0.1796}$(22)	$0.3502_{\pm 0.0284}$(23)
LMDD	$0.7921_{\pm 0.1227}$(18)	$0.0951_{\pm 0.0897}$(24)	$0.2530_{\pm 0.1153}$(22)	$0.5672_{\pm 0.0604}$(20)	$0.7597_{\pm 0.1644}$(19)	$0.4895_{\pm 0.0585}$(16)
LOF	$0.3152_{\pm 0.0156}$(21)	$0.2320_{\pm 0.1363}$(8)	$0.4741_{\pm 0.1082}$(16)	$0.8282_{\pm 0.0611}$(16)	$0.9833_{\pm 0.0351}$(7)	$0.4409_{\pm 0.0227}$(20)
LUNAR	$0.9798_{\pm 0.0101}$(12)	$0.2542_{\pm 0.1299}$(4)	$0.4797_{\pm 0.1632}$(15)	$0.9677_{\pm 0.0119}$(**1**)	$0.8629_{\pm 0.1506}$(15)	$0.5502_{\pm 0.0212}$(5)
MCD	$0.9630_{\pm 0.0222}$(16)	$0.1475_{\pm 0.0490}$(19)	$0.4847_{\pm 0.1319}$(14)	$0.9534_{\pm 0.0115}$(4)	$0.8314_{\pm 0.1602}$(16)	$0.5054_{\pm 0.0295}$(12)
MO_GAAL	$0.2116_{\pm 0.0082}$(24)	$0.1409_{\pm 0.1880}$(21)	$0.2080_{\pm 0.1225}$(24)	$0.2876_{\pm 0.0459}$(24)	$0.0339_{\pm 0.0072}$(24)	$0.2784_{\pm 0.0480}$(25)
OCSVM	$0.9904_{\pm 0.0041}$(3)	$0.1249_{\pm 0.1108}$(23)	$0.5231_{\pm 0.1038}$(8)	$0.8001_{\pm 0.0329}$(17)	$0.9833_{\pm 0.0351}$(7)	$0.5028_{\pm 0.0288}$(13)
PCA	$0.9507_{\pm 0.0178}$(17)	$0.1454_{\pm 0.1055}$(20)	$0.4853_{\pm 0.1374}$(13)	$0.7236_{\pm 0.0393}$(18)	$0.9333_{\pm 0.1291}$(11)	$0.4678_{\pm 0.0199}$(19)
QMCD	$0.5168_{\pm 0.1374}$(20)	$0.2025_{\pm 0.0687}$(11)	$0.3999_{\pm 0.1074}$(20)	$0.3946_{\pm 0.0602}$(22)	$0.0378_{\pm 0.0084}$(23)	$0.6099_{\pm 0.0347}$(**1**)
SO_GAAL	$0.2116_{\pm 0.0082}$(25)	$0.1298_{\pm 0.1725}$(22)	$0.1719_{\pm 0.0960}$(25)	$0.2702_{\pm 0.0379}$(25)	$0.0338_{\pm 0.0071}$(25)	$0.2784_{\pm 0.0450}$(24)
Sampling	$0.9878_{\pm 0.0061}$(6)	$0.2452_{\pm 0.1173}$(7)	$0.4322_{\pm 0.1248}$(18)	$0.9093_{\pm 0.0388}$(12)	$0.8901_{\pm 0.1443}$(13)	$0.5312_{\pm 0.0416}$(8)
TCCM	$0.9718_{\pm 0.0148}$(13)	$0.2536_{\pm 0.1251}$(5)	$0.5267_{\pm 0.0975}$(7)	$0.9071_{\pm 0.0227}$(13)	$1.0000_{\pm 0.0000}$(**1**)	$0.4202_{\pm 0.0638}$(21)
VAE	$0.9806_{\pm 0.0108}$(9)	$0.1503_{\pm 0.0995}$(17)	$0.6372_{\pm 0.1153}$(**1**)	$0.8817_{\pm 0.0283}$(14)	$0.8796_{\pm 0.1430}$(14)	$0.4876_{\pm 0.0228}$(17)

TABLE XXI: AUPRC, small datasets (2/2).

Model	37_Stamps	39_vertebral	42_WBC	43_WDBC	45_wine	46_WPBC
ABOD	$0.2120_{\pm 0.0573}$(21)	$0.1222_{\pm 0.0268}$(19)	$0.3421_{\pm 0.1683}$(20)	$0.2573_{\pm 0.1460}$(20)	$0.1717_{\pm 0.0927}$(22)	$0.2141_{\pm 0.0518}$(24)
AutoEncoder	$0.4350_{\pm 0.1051}$(15)	$0.1265_{\pm 0.0253}$(16)	$0.8106_{\pm 0.2272}$(15)	$0.7782_{\pm 0.2134}$(5)	$0.5529_{\pm 0.2822}$(6)	$0.2347_{\pm 0.0631}$(14)
CBLOF	$0.5528_{\pm 0.1300}$(4)	$0.1314_{\pm 0.0278}$(11)	$0.9480_{\pm 0.0504}$(5)	$0.6179_{\pm 0.2237}$(15)	$0.6318_{\pm 0.2372}$(2)	$0.2413_{\pm 0.0610}$(8)
DIF	$0.5181_{\pm 0.0937}$(6)	$0.1855_{\pm 0.0487}$(3)	$0.1239_{\pm 0.0324}$(21)	$0.0662_{\pm 0.0339}$(22)	$0.2498_{\pm 0.1465}$(20)	$0.2223_{\pm 0.0594}$(21)
DiffInt-DA	$0.6055_{\pm 0.0710}$(2)	$0.1571_{\pm 0.0154}$(6)	$0.9044_{\pm 0.0522}$(10)	$0.9828_{\pm 0.0338}$(2)	$0.4242_{\pm 0.0568}$(15)	$0.3048_{\pm 0.0349}$(**1**)
DiffInt-IA	$0.6033_{\pm 0.0703}$(3)	$0.1626_{\pm 0.0173}$(5)	$0.9107_{\pm 0.0751}$(9)	$0.9958_{\pm 0.0328}$(**1**)	$0.5478_{\pm 0.1085}$(7)	$0.2949_{\pm 0.0402}$(2)
DiffInt-PA	$0.6095_{\pm 0.0647}$(**1**)	$0.1548_{\pm 0.0192}$(7)	$0.9658_{\pm 0.0466}$(**1**)	$0.5729_{\pm 0.1161}$(18)	$0.4625_{\pm 0.1262}$(12)	$0.2834_{\pm 0.0336}$(3)
ECOD	$0.4575_{\pm 0.1030}$(13)	$0.1310_{\pm 0.0239}$(12)	$0.9174_{\pm 0.0831}$(8)	$0.6875_{\pm 0.1531}$(8)	$0.3033_{\pm 0.0885}$(18)	$0.2158_{\pm 0.0474}$(23)
GMM	$0.4618_{\pm 0.0950}$(12)	$0.1291_{\pm 0.0219}$(14)	$0.6966_{\pm 0.1988}$(19)	$0.6851_{\pm 0.2442}$(10)	$0.6901_{\pm 0.2396}$(**1**)	$0.2392_{\pm 0.0704}$(10)
HBOS	$0.4570_{\pm 0.0663}$(14)	$0.1110_{\pm 0.0243}$(23)	$0.8305_{\pm 0.1530}$(14)	$0.7374_{\pm 0.1368}$(6)	$0.5273_{\pm 0.2059}$(10)	$0.2355_{\pm 0.0603}$(13)
IForest	$0.5178_{\pm 0.1048}$(7)	$0.1224_{\pm 0.0263}$(18)	$0.9201_{\pm 0.0784}$(7)	$0.6873_{\pm 0.1773}$(9)	$0.5717_{\pm 0.1853}$(5)	$0.2344_{\pm 0.0597}$(15)
KDE	$0.4759_{\pm 0.0940}$(10)	$0.1144_{\pm 0.0241}$(22)	$0.9502_{\pm 0.0568}$(4)	$0.6224_{\pm 0.1666}$(14)	$0.5357_{\pm 0.1809}$(8)	$0.2503_{\pm 0.0598}$(7)
KPCA	$0.0939_{\pm 0.0238}$(23)	$0.1707_{\pm 0.0392}$(4)	$0.1196_{\pm 0.1020}$(23)	$0.0208_{\pm 0.0054}$(23)	$0.1054_{\pm 0.0808}$(23)	$0.2364_{\pm 0.0843}$(12)
LMDD	$0.1085_{\pm 0.0399}$(22)	$0.1233_{\pm 0.0291}$(17)	$0.9302_{\pm 0.0867}$(6)	$0.7950_{\pm 0.1495}$(3)	$0.2445_{\pm 0.1397}$(21)	$0.2394_{\pm 0.0452}$(9)
LOF	$0.2330_{\pm 0.0598}$(20)	$0.1073_{\pm 0.0211}$(25)	$0.7069_{\pm 0.2313}$(18)	$0.6436_{\pm 0.1994}$(12)	$0.4505_{\pm 0.1887}$(13)	$0.2377_{\pm 0.0612}$(11)
LUNAR	$0.5381_{\pm 0.1321}$(5)	$0.1283_{\pm 0.0321}$(15)	$0.7695_{\pm 0.2384}$(17)	$0.7913_{\pm 0.1626}$(4)	$0.6064_{\pm 0.2519}$(3)	$0.2513_{\pm 0.0619}$(6)
MCD	$0.2714_{\pm 0.0547}$(19)	$0.1342_{\pm 0.0274}$(10)	$0.7814_{\pm 0.2436}$(16)	$0.4138_{\pm 0.1816}$(19)	$0.5772_{\pm 0.2406}$(4)	$0.2583_{\pm 0.0910}$(5)
MO_GAAL	$0.0723_{\pm 0.0240}$(24)	$0.2425_{\pm 0.0945}$(**1**)	$0.0282_{\pm 0.0047}$(25)	$0.0158_{\pm 0.0044}$(24)	$0.0440_{\pm 0.0147}$(24)	$0.2211_{\pm 0.0706}$(22)
OCSVM	$0.4249_{\pm 0.1001}$(16)	$0.1202_{\pm 0.0253}$(20)	$0.9502_{\pm 0.0568}$(3)	$0.6429_{\pm 0.1641}$(13)	$0.3484_{\pm 0.1777}$(16)	$0.2333_{\pm 0.0561}$(16)
PCA	$0.4083_{\pm 0.0799}$(17)	$0.1093_{\pm 0.0161}$(24)	$0.9009_{\pm 0.0873}$(11)	$0.6025_{\pm 0.1176}$(16)	$0.3114_{\pm 0.1194}$(17)	$0.2276_{\pm 0.0529}$(20)
QMCD	$0.4661_{\pm 0.0532}$(11)	$0.1144_{\pm 0.0252}$(21)	$0.1227_{\pm 0.0592}$(22)	$0.2214_{\pm 0.1521}$(21)	$0.2567_{\pm 0.1461}$(19)	$0.2330_{\pm 0.0558}$(17)
SO_GAAL	$0.0658_{\pm 0.0152}$(25)	$0.2141_{\pm 0.0595}$(2)	$0.0282_{\pm 0.0047}$(24)	$0.0158_{\pm 0.0044}$(25)	$0.0438_{\pm 0.0149}$(25)	$0.2100_{\pm 0.0529}$(25)
Sampling	$0.4803_{\pm 0.1574}$(9)	$0.1291_{\pm 0.0358}$(13)	$0.8885_{\pm 0.2170}$(12)	$0.5865_{\pm 0.1612}$(17)	$0.5280_{\pm 0.1743}$(9)	$0.2612_{\pm 0.0817}$(4)
TCCM	$0.3988_{\pm 0.0925}$(18)	$0.1363_{\pm 0.0268}$(9)	$0.9610_{\pm 0.0576}$(2)	$0.6813_{\pm 0.1950}$(11)	$0.5198_{\pm 0.2062}$(11)	$0.2295_{\pm 0.0547}$(18)
VAE	$0.4855_{\pm 0.0914}$(8)	$0.1434_{\pm 0.0283}$(8)	$0.8563_{\pm 0.1843}$(13)	$0.6896_{\pm 0.1968}$(7)	$0.4389_{\pm 0.1942}$(14)	$0.2283_{\pm 0.0544}$(19)

TABLE XXII: AUPRC, medium datasets (1/3).

Model	2_annthyroid	6_cardio	7_Cardiotocography	12_fault	19_landsat	20_letter
ABOD	$0.1685_{\pm0.0118}$(19)	$0.1865_{\pm0.0217}$(21)	$0.2591_{\pm0.0182}$(22)	$0.5010_{\pm0.0160}$(12)	$0.2219_{\pm0.0116}$(20)	$0.3120_{\pm0.0422}$(7)
AutoEncoder	$0.5003_{\pm0.0592}$(4)	$0.6195_{\pm0.0650}$(14)	$0.4920_{\pm0.0476}$(15)	$0.5937_{\pm0.0420}$(7)	$0.2604_{\pm0.0116}$(12)	$0.2659_{\pm0.0448}$(9)
CBLOF	$0.1605_{\pm0.0144}$(20)	$0.7159_{\pm0.0405}$(9)	$0.5803_{\pm0.0271}$(7)	$0.5318_{\pm0.0543}$(11)	$0.3355_{\pm0.0152}$(6)	$0.1951_{\pm0.0339}$(11)
DIF	$0.4211_{\pm0.0358}$(9)	$0.7767_{\pm0.0339}$(4)	$0.5001_{\pm0.0246}$(14)	$0.5894_{\pm0.0342}$(8)	$0.2932_{\pm0.0103}$(9)	$0.1555_{\pm0.0476}$(12)
DiffInt-DA	$0.4439_{\pm0.0248}$(7)	$0.7385_{\pm0.0380}$(6)	$0.5714_{\pm0.0227}$(8)	$0.6212_{\pm0.0137}$(5)	$0.5041_{\pm0.0109}$(3)	$0.2762_{\pm0.0218}$(8)
DiffInt-IA	$0.3593_{\pm0.0172}$(12)	$0.7359_{\pm0.0314}$(7)	$0.5355_{\pm0.0177}$(10)	$0.6313_{\pm0.0146}$(3)	$0.5291_{\pm0.0117}$(**1**)	$0.4655_{\pm0.0368}$(2)
DiffInt-PA	$0.3652_{\pm0.0133}$(11)	$0.8114_{\pm0.0213}$(2)	$0.6271_{\pm0.0166}$(**1**)	$0.6639_{\pm0.0128}$(2)	$0.3688_{\pm0.0088}$(4)	$0.3954_{\pm0.0306}$(3)
ECOD	$0.3479_{\pm0.0222}$(13)	$0.6705_{\pm0.0231}$(13)	$0.5820_{\pm0.0238}$(6)	$0.3467_{\pm0.0165}$(24)	$0.1769_{\pm0.0043}$(23)	$0.0899_{\pm0.0180}$(18)
GMM	$0.4365_{\pm0.0291}$(8)	$0.6968_{\pm0.0376}$(10)	$0.5269_{\pm0.0250}$(11)	$0.5459_{\pm0.0311}$(9)	$0.2060_{\pm0.0064}$(22)	$0.3384_{\pm0.0526}$(4)
HBOS	$0.3934_{\pm0.0471}$(10)	$0.5475_{\pm0.0316}$(16)	$0.4559_{\pm0.0241}$(16)	$0.4442_{\pm0.0669}$(17)	$0.3580_{\pm0.0178}$(5)	$0.0867_{\pm0.0190}$(20)
IForest	$0.4849_{\pm0.0345}$(5)	$0.6925_{\pm0.0521}$(11)	$0.5648_{\pm0.0281}$(9)	$0.4958_{\pm0.0382}$(13)	$0.2668_{\pm0.0149}$(11)	$0.1011_{\pm0.0206}$(17)
KDE	$0.1274_{\pm0.0129}$(22)	$0.7267_{\pm0.0282}$(8)	$0.6152_{\pm0.0210}$(3)	$0.6196_{\pm0.0279}$(6)	$0.2385_{\pm0.0055}$(16)	$0.1506_{\pm0.0325}$(13)
KPCA	$0.2036_{\pm0.0217}$(15)	$0.0928_{\pm0.0123}$(24)	$0.2136_{\pm0.0213}$(23)	$0.3633_{\pm0.0188}$(21)	$0.2150_{\pm0.0096}$(21)	$0.0714_{\pm0.0089}$(22)
LMDD	$0.1165_{\pm0.0131}$(23)	$0.1569_{\pm0.0189}$(22)	$0.4269_{\pm0.0372}$(17)	$0.3829_{\pm0.0243}$(20)	$0.2274_{\pm0.0884}$(19)	$0.0638_{\pm0.0079}$(23)
LOF	$0.1901_{\pm0.0185}$(17)	$0.2026_{\pm0.0235}$(20)	$0.2883_{\pm0.0292}$(20)	$0.4002_{\pm0.0206}$(19)	$0.2457_{\pm0.0095}$(15)	$0.3357_{\pm0.0444}$(5)
LUNAR	$0.4683_{\pm0.0401}$(6)	$0.7791_{\pm0.0356}$(3)	$0.6128_{\pm0.0307}$(4)	$0.6845_{\pm0.0358}$(**1**)	$0.5202_{\pm0.0228}$(2)	$0.5013_{\pm0.0652}$(**1**)
MCD	$0.5152_{\pm0.0299}$(2)	$0.3805_{\pm0.0375}$(17)	$0.3595_{\pm0.0297}$(18)	$0.4347_{\pm0.0468}$(18)	$0.2593_{\pm0.0042}$(13)	$0.1998_{\pm0.0415}$(10)
MO_GAAL	$0.1013_{\pm0.0212}$(24)	$0.0918_{\pm0.0954}$(25)	$0.2120_{\pm0.0732}$(24)	$0.3535_{\pm0.0403}$(22)	$0.3300_{\pm0.0331}$(7)	$0.0577_{\pm0.0097}$(24)
OCSVM	$0.1313_{\pm0.0133}$(21)	$0.6923_{\pm0.0300}$(12)	$0.6186_{\pm0.0240}$(2)	$0.4703_{\pm0.0216}$(15)	$0.1712_{\pm0.0046}$(24)	$0.0807_{\pm0.0169}$(21)
PCA	$0.2015_{\pm0.0113}$(16)	$0.3667_{\pm0.2827}$(18)	$0.2990_{\pm0.1234}$(19)	$0.3415_{\pm0.0173}$(25)	$0.1635_{\pm0.0041}$(25)	$0.0878_{\pm0.0205}$(19)
QMCD	$0.2606_{\pm0.0275}$(14)	$0.3538_{\pm0.0571}$(19)	$0.2713_{\pm0.0218}$(21)	$0.4705_{\pm0.0492}$(14)	$0.3152_{\pm0.0113}$(8)	$0.1023_{\pm0.0231}$(16)
SO_GAAL	$0.0981_{\pm0.0196}$(25)	$0.0934_{\pm0.0961}$(23)	$0.2083_{\pm0.0727}$(25)	$0.3521_{\pm0.0374}$(23)	$0.2349_{\pm0.0285}$(18)	$0.0559_{\pm0.0080}$(25)
Sampling	$0.1741_{\pm0.0209}$(18)	$0.6043_{\pm0.0843}$(15)	$0.5187_{\pm0.0438}$(13)	$0.5323_{\pm0.0531}$(10)	$0.2839_{\pm0.0537}$(10)	$0.1489_{\pm0.0309}$(14)
TCCM	$0.5279_{\pm0.0935}$(**1**)	$0.8352_{\pm0.0315}$(**1**)	$0.5254_{\pm0.0509}$(12)	$0.6234_{\pm0.0299}$(4)	$0.2533_{\pm0.0126}$(14)	$0.3177_{\pm0.0507}$(6)
VAE	$0.5096_{\pm0.0311}$(3)	$0.7723_{\pm0.0403}$(5)	$0.6065_{\pm0.0273}$(5)	$0.4672_{\pm0.0354}$(16)	$0.2372_{\pm0.0057}$(17)	$0.1301_{\pm0.0299}$(15)

TABLE XXIII: AUPRC, medium datasets (2/3).

Model	27_PageBlocks	28_pendigits	30_satellite	31_satimage-2	38_thyroid	40_vowels
ABOD	$0.3540_{\pm0.0282}$(20)	$0.0601_{\pm0.0106}$(20)	$0.3867_{\pm0.0150}$(22)	$0.1329_{\pm0.0474}$(20)	$0.1251_{\pm0.0139}$(23)	$0.4213_{\pm0.1150}$(9)
AutoEncoder	$0.7179_{\pm0.0330}$(5)	$0.4177_{\pm0.0942}$(9)	$0.7908_{\pm0.0112}$(8)	$0.8612_{\pm0.0970}$(14)	$0.6788_{\pm0.0687}$(5)	$0.5336_{\pm0.0721}$(5)
CBLOF	$0.5343_{\pm0.0463}$(14)	$0.4306_{\pm0.1145}$(8)	$0.7996_{\pm0.0198}$(7)	$0.9727_{\pm0.0222}$(5)	$0.2328_{\pm0.0654}$(18)	$0.3702_{\pm0.1387}$(10)
DIF	$0.6854_{\pm0.0315}$(8)	$0.5697_{\pm0.1402}$(7)	$0.7803_{\pm0.0134}$(9)	$0.9036_{\pm0.0684}$(11)	$0.7491_{\pm0.0391}$(**1**)	$0.2130_{\pm0.0693}$(12)
DiffInt-DA	$0.7266_{\pm0.0136}$(3)	$0.8625_{\pm0.0560}$(2)	$0.8149_{\pm0.0057}$(4)	$0.9888_{\pm0.0071}$(**1**)	$0.5781_{\pm0.0479}$(10)	$0.4968_{\pm0.0516}$(6)
DiffInt-IA	$0.7132_{\pm0.0262}$(6)	$0.8069_{\pm0.0605}$(4)	$0.8278_{\pm0.0071}$(2)	$0.9782_{\pm0.0117}$(2)	$0.5157_{\pm0.0467}$(12)	$0.6080_{\pm0.0552}$(4)
DiffInt-PA	$0.7673_{\pm0.0121}$(**1**)	$0.8617_{\pm0.0325}$(3)	$0.8168_{\pm0.0060}$(3)	$0.9727_{\pm0.0117}$(4)	$0.5704_{\pm0.0412}$(11)	$0.7692_{\pm0.0502}$(2)
ECOD	$0.5734_{\pm0.0258}$(11)	$0.3003_{\pm0.0431}$(13)	$0.5882_{\pm0.0081}$(19)	$0.7695_{\pm0.0614}$(17)	$0.6398_{\pm0.0425}$(8)	$0.1090_{\pm0.0263}$(18)
GMM	$0.7209_{\pm0.0241}$(4)	$0.0879_{\pm0.0160}$(18)	$0.7768_{\pm0.0111}$(10)	$0.7337_{\pm0.1016}$(18)	$0.6474_{\pm0.0461}$(7)	$0.6293_{\pm0.0983}$(3)
HBOS	$0.3019_{\pm0.0160}$(21)	$0.2869_{\pm0.0357}$(14)	$0.8109_{\pm0.0092}$(6)	$0.8270_{\pm0.0647}$(15)	$0.7392_{\pm0.0772}$(3)	$0.1337_{\pm0.0498}$(15)
IForest	$0.5398_{\pm0.0328}$(13)	$0.3881_{\pm0.0638}$(10)	$0.7662_{\pm0.0140}$(12)	$0.9210_{\pm0.0427}$(9)	$0.6635_{\pm0.0942}$(6)	$0.1938_{\pm0.0504}$(13)
KDE	$0.4864_{\pm0.0316}$(17)	$0.7551_{\pm0.0752}$(5)	$0.7479_{\pm0.0115}$(14)	$0.9771_{\pm0.0245}$(3)	$0.1606_{\pm0.0532}$(22)	$0.1787_{\pm0.0573}$(14)
KPCA	$0.1674_{\pm0.0129}$(23)	$0.0228_{\pm0.0061}$(23)	$0.3056_{\pm0.0142}$(24)	$0.0628_{\pm0.1632}$(22)	$0.0273_{\pm0.0118}$(25)	$0.0378_{\pm0.0092}$(23)
LMDD	$0.4825_{\pm0.0221}$(18)	$0.0283_{\pm0.0061}$(22)	$0.3157_{\pm0.0782}$(23)	$0.0212_{\pm0.0224}$(24)	$0.0696_{\pm0.0161}$(24)	$0.0498_{\pm0.0255}$(22)
LOF	$0.4083_{\pm0.0256}$(19)	$0.0376_{\pm0.0119}$(21)	$0.3932_{\pm0.0145}$(21)	$0.0305_{\pm0.0175}$(23)	$0.2889_{\pm0.0935}$(15)	$0.4563_{\pm0.0951}$(8)
LUNAR	$0.7430_{\pm0.0304}$(2)	$0.9659_{\pm0.0331}$(**1**)	$0.8405_{\pm0.0110}$(**1**)	$0.9656_{\pm0.0268}$(7)	$0.6089_{\pm0.0850}$(9)	$0.8212_{\pm0.0922}$(**1**)
MCD	$0.6209_{\pm0.0226}$(10)	$0.0722_{\pm0.0102}$(19)	$0.7754_{\pm0.0136}$(11)	$0.7194_{\pm0.0769}$(19)	$0.7096_{\pm0.0493}$(4)	$0.0977_{\pm0.0686}$(19)
MO_GAAL	$0.1085_{\pm0.0568}$(25)	$0.0134_{\pm0.0016}$(24)	$0.2918_{\pm0.0194}$(25)	$0.0080_{\pm0.0029}$(25)	$0.2146_{\pm0.1110}$(19)	$0.0191_{\pm0.0025}$(24)
OCSVM	$0.5083_{\pm0.0317}$(16)	$0.3706_{\pm0.0533}$(11)	$0.6837_{\pm0.0107}$(17)	$0.9066_{\pm0.0385}$(10)	$0.1701_{\pm0.0539}$(21)	$0.0954_{\pm0.0293}$(20)
PCA	$0.5327_{\pm0.0308}$(15)	$0.2323_{\pm0.0378}$(16)	$0.6038_{\pm0.0099}$(18)	$0.8750_{\pm0.0433}$(13)	$0.3753_{\pm0.0389}$(13)	$0.0816_{\pm0.0254}$(21)
QMCD	$0.2177_{\pm0.0127}$(22)	$0.1684_{\pm0.0255}$(17)	$0.8145_{\pm0.0096}$(5)	$0.9346_{\pm0.0343}$(8)	$0.3499_{\pm0.0531}$(14)	$0.1307_{\pm0.0329}$(16)
SO_GAAL	$0.1109_{\pm0.0540}$(24)	$0.0134_{\pm0.0013}$(25)	$0.4639_{\pm0.1478}$(20)	$0.1315_{\pm0.2360}$(21)	$0.2050_{\pm0.0953}$(20)	$0.0191_{\pm0.0027}$(25)
Sampling	$0.5436_{\pm0.0792}$(12)	$0.3038_{\pm0.2087}$(12)	$0.7445_{\pm0.0386}$(15)	$0.8896_{\pm0.1368}$(12)	$0.2644_{\pm0.0744}$(17)	$0.3241_{\pm0.0924}$(11)
TCCM	$0.7055_{\pm0.0705}$(7)	$0.6903_{\pm0.1344}$(6)	$0.7586_{\pm0.0199}$(13)	$0.9695_{\pm0.0281}$(6)	$0.2650_{\pm0.0576}$(16)	$0.4667_{\pm0.1105}$(7)
VAE	$0.6740_{\pm0.0416}$(9)	$0.2768_{\pm0.0448}$(15)	$0.7325_{\pm0.0113}$(16)	$0.8211_{\pm0.0622}$(16)	$0.7433_{\pm0.0486}$(2)	$0.1170_{\pm0.0567}$(17)

TABLE XXIV: AUPRC, medium datasets (3/3).

Model	41_Waveform	44_Wilt	47_yeast
ABOD	0.0432$_{\pm0.0084}$(17)	0.0548$_{\pm0.0024}$(11)	0.3089$_{\pm0.0129}$(21)
AutoEncoder	0.0787$_{\pm0.0299}$(7)	0.0562$_{\pm0.0139}$(10)	0.3197$_{\pm0.0182}$(15)
CBLOF	0.2284$_{\pm0.0701}$(**1**)	0.0387$_{\pm0.0030}$(22)	0.3223$_{\pm0.0202}$(13)
DIF	0.0873$_{\pm0.0246}$(6)	0.0419$_{\pm0.0059}$(20)	0.2961$_{\pm0.0133}$(25)
DiffInt-DA	0.0505$_{\pm0.0066}$(15)	0.1964$_{\pm0.0408}$(2)	0.3403$_{\pm0.0094}$(7)
DiffInt-IA	0.0602$_{\pm0.0067}$(11)	0.2342$_{\pm0.0172}$(**1**)	0.3410$_{\pm0.0100}$(6)
DiffInt-PA	0.0921$_{\pm0.0117}$(5)	0.0798$_{\pm0.0060}$(7)	0.3534$_{\pm0.0129}$(5)
ECOD	0.0431$_{\pm0.0086}$(18)	0.0428$_{\pm0.0048}$(17)	0.3393$_{\pm0.0157}$(8)
GMM	0.0426$_{\pm0.0086}$(20)	0.0918$_{\pm0.0082}$(5)	0.3096$_{\pm0.0171}$(19)
HBOS	0.0508$_{\pm0.0073}$(14)	0.0444$_{\pm0.0094}$(15)	0.3334$_{\pm0.0165}$(10)
IForest	0.0626$_{\pm0.0121}$(9)	0.0453$_{\pm0.0033}$(14)	0.3096$_{\pm0.0167}$(18)
KDE	0.0696$_{\pm0.0097}$(8)	0.0385$_{\pm0.0030}$(24)	0.2974$_{\pm0.0143}$(23)
KPCA	0.0320$_{\pm0.0048}$(23)	0.0717$_{\pm0.0053}$(8)	0.3807$_{\pm0.0174}$(**1**)
LMDD	0.0323$_{\pm0.0044}$(22)	0.0494$_{\pm0.0070}$(12)	0.3337$_{\pm0.0185}$(9)
LOF	0.1167$_{\pm0.0347}$(4)	0.0426$_{\pm0.0029}$(18)	0.3091$_{\pm0.0220}$(20)
LUNAR	0.2161$_{\pm0.0711}$(2)	0.0612$_{\pm0.0128}$(9)	0.3226$_{\pm0.0206}$(12)
MCD	0.0427$_{\pm0.0091}$(19)	0.1547$_{\pm0.0106}$(3)	0.3211$_{\pm0.0242}$(14)
MO_GAAL	0.0212$_{\pm0.0064}$(25)	0.0928$_{\pm0.0121}$(4)	0.3647$_{\pm0.0406}$(3)
OCSVM	0.0398$_{\pm0.0081}$(21)	0.0387$_{\pm0.0032}$(21)	0.3063$_{\pm0.0138}$(22)
PCA	0.0466$_{\pm0.0082}$(16)	0.0432$_{\pm0.0026}$(16)	0.2967$_{\pm0.0169}$(24)
QMCD	0.0594$_{\pm0.0072}$(12)	0.0377$_{\pm0.0032}$(25)	0.3283$_{\pm0.0192}$(11)
SO_GAAL	0.0237$_{\pm0.0083}$(24)	0.0902$_{\pm0.0115}$(6)	0.3660$_{\pm0.0508}$(2)
Sampling	0.1238$_{\pm0.0543}$(3)	0.0420$_{\pm0.0030}$(19)	0.3179$_{\pm0.0256}$(16)
TCCM	0.0562$_{\pm0.0186}$(13)	0.0385$_{\pm0.0049}$(23)	0.3585$_{\pm0.0255}$(4)
VAE	0.0616$_{\pm0.0121}$(10)	0.0462$_{\pm0.0036}$(13)	0.3138$_{\pm0.0179}$(17)

TABLE XXV: AUPRC, large datasets (1/2).

Model	1_ALOI	8_celeba	10_cover	11_donors	13_fraud	16_http
ABOD	0.1185$_{\pm0.0111}$(2)	0.0202$_{\pm0.0005}$(21)	0.0288$_{\pm0.0013}$(19)	0.0492$_{\pm0.0003}$(21)	0.0159$_{\pm0.0011}$(20)	0.0077$_{\pm0.0002}$(20)
AutoEncoder	0.0419$_{\pm0.0033}$(10)	0.0499$_{\pm0.0104}$(12)	0.4044$_{\pm0.0611}$(5)	0.3215$_{\pm0.0309}$(6)	0.5606$_{\pm0.0488}$(3)	0.7764$_{\pm0.1551}$(8)
CBLOF	0.0406$_{\pm0.0034}$(12)	0.0617$_{\pm0.0294}$(10)	0.0772$_{\pm0.0259}$(12)	0.3004$_{\pm0.0451}$(8)	0.4031$_{\pm0.0301}$(10)	0.4121$_{\pm0.0062}$(15)
DIF	0.0438$_{\pm0.0026}$(7)	0.0491$_{\pm0.0041}$(14)	0.3492$_{\pm0.0373}$(6)	0.2382$_{\pm0.0192}$(11)	0.4106$_{\pm0.0472}$(9)	0.3531$_{\pm0.0072}$(17)
DiffInt-DA	0.0681$_{\pm0.0031}$(6)	0.0386$_{\pm0.0061}$(17)	0.7806$_{\pm0.0416}$(3)	0.9990$_{\pm0.0004}$(3)	0.6153$_{\pm0.0210}$(2)	0.9455$_{\pm0.0831}$(2)
DiffInt-IA	0.0685$_{\pm0.0031}$(5)	0.0493$_{\pm0.0049}$(13)	0.7572$_{\pm0.0680}$(4)	0.9961$_{\pm0.0007}$(4)	0.6315$_{\pm0.0227}$(**1**)	0.9151$_{\pm0.0954}$(3)
DiffInt-PA	0.0807$_{\pm0.0033}$(4)	0.0528$_{\pm0.0056}$(11)	0.8938$_{\pm0.0073}$(**1**)	0.9994$_{\pm0.0005}$(**1**)	0.5503$_{\pm0.0193}$(4)	0.8872$_{\pm0.1252}$(4)
ECOD	0.0335$_{\pm0.0014}$(22)	0.1007$_{\pm0.0055}$(4)	0.1419$_{\pm0.0065}$(9)	0.3098$_{\pm0.0022}$(7)	0.2513$_{\pm0.0266}$(15)	0.2311$_{\pm0.0028}$(18)
GMM	0.0403$_{\pm0.0025}$(13)	0.0910$_{\pm0.0042}$(6)	0.1114$_{\pm0.0025}$(10)	0.2779$_{\pm0.0016}$(10)	0.4888$_{\pm0.0380}$(6)	0.8511$_{\pm0.0052}$(7)
HBOS	0.0343$_{\pm0.0013}$(19)	0.1009$_{\pm0.0056}$(3)	0.0352$_{\pm0.0019}$(18)	0.1826$_{\pm0.0026}$(16)	0.2132$_{\pm0.0351}$(16)	0.5106$_{\pm0.1107}$(11)
IForest	0.0337$_{\pm0.0016}$(21)	0.0682$_{\pm0.0095}$(9)	0.0384$_{\pm0.0049}$(16)	0.2907$_{\pm0.0495}$(9)	0.1265$_{\pm0.0279}$(19)	0.3906$_{\pm0.0891}$(16)
KDE	0.0423$_{\pm0.0034}$(8)	0.0406$_{\pm0.0021}$(16)	0.0495$_{\pm0.0016}$(14)	-	0.3945$_{\pm0.0308}$(13)	-
KPCA	0.0386$_{\pm0.0017}$(16)	-	-	-	-	-
LMDD	0.0364$_{\pm0.0017}$(18)	-	-	-	-	-
LOF	0.0901$_{\pm0.0074}$(3)	0.0172$_{\pm0.0006}$(22)	0.0136$_{\pm0.0012}$(21)	0.1053$_{\pm0.0021}$(18)	0.0018$_{\pm0.0001}$(21)	0.0254$_{\pm0.0038}$(19)
LUNAR	0.1491$_{\pm0.0110}$(**1**)	0.0356$_{\pm0.0018}$(18)	0.8678$_{\pm0.0130}$(2)	0.9991$_{\pm0.0006}$(2)	0.4674$_{\pm0.0555}$(7)	0.5629$_{\pm0.3048}$(9)
MCD	0.0341$_{\pm0.0017}$(20)	0.0958$_{\pm0.0171}$(5)	0.0158$_{\pm0.0004}$(20)	0.1983$_{\pm0.1187}$(14)	0.5063$_{\pm0.0249}$(5)	0.8727$_{\pm0.0061}$(5)
MO_GAAL	0.0290$_{\pm0.0016}$(25)	0.0136$_{\pm0.0017}$(23)	0.0101$_{\pm0.0008}$(22)	-	0.0009$_{\pm0.0000}$(23)	-
OCSVM	0.0421$_{\pm0.0034}$(9)	0.0772$_{\pm0.0048}$(8)	0.1093$_{\pm0.0046}$(11)	0.2029$_{\pm0.0021}$(13)	0.3949$_{\pm0.0306}$(12)	0.4138$_{\pm0.0067}$(14)
PCA	0.0380$_{\pm0.0018}$(17)	0.1121$_{\pm0.0050}$(2)	0.0732$_{\pm0.0015}$(13)	0.1673$_{\pm0.0012}$(17)	0.1506$_{\pm0.0159}$(18)	0.4758$_{\pm0.0056}$(12)
QMCD	0.0322$_{\pm0.0011}$(23)	0.0222$_{\pm0.0006}$(19)	0.0378$_{\pm0.0013}$(17)	0.0937$_{\pm0.0010}$(19)	0.2982$_{\pm0.0382}$(14)	0.5394$_{\pm0.0175}$(10)
SO_GAAL	0.0292$_{\pm0.0015}$(24)	0.0214$_{\pm0.0010}$(20)	0.0100$_{\pm0.0003}$(23)	0.0513$_{\pm0.0082}$(20)	0.0009$_{\pm0.0001}$(22)	0.0038$_{\pm0.0001}$(21)
Sampling	0.0400$_{\pm0.0036}$(14)	0.0432$_{\pm0.0264}$(15)	0.0465$_{\pm0.0212}$(15)	0.2270$_{\pm0.0496}$(12)	0.3992$_{\pm0.0286}$(11)	0.4480$_{\pm0.1252}$(13)
TCCM	0.0406$_{\pm0.0032}$(11)	0.0833$_{\pm0.0067}$(7)	0.2212$_{\pm0.1091}$(7)	0.9181$_{\pm0.0839}$(5)	0.4241$_{\pm0.0526}$(8)	0.9790$_{\pm0.0097}$(**1**)
VAE	0.0393$_{\pm0.0021}$(15)	0.1142$_{\pm0.0058}$(**1**)	0.1664$_{\pm0.0069}$(8)	0.1888$_{\pm0.0413}$(15)	0.1761$_{\pm0.0192}$(17)	0.8511$_{\pm0.0060}$(6)

TABLE XXVI: AUPRC, large datasets (2/2).

Model	22_magic	23_mammography	32_shuttle	33_skin	34_smtp
ABOD	$0.7128_{\pm 0.0079}$(11)	$0.0240_{\pm 0.0016}$(23)	$0.1325_{\pm 0.0051}$(21)	$0.1771_{\pm 0.0013}$(23)	$0.0014_{\pm 0.0002}$(22)
AutoEncoder	$0.7687_{\pm 0.0135}$(5)	$0.3620_{\pm 0.0625}$(2)	$0.9496_{\pm 0.0117}$(14)	$0.4034_{\pm 0.1190}$(12)	$0.3247_{\pm 0.2130}$(12)
CBLOF	$0.7596_{\pm 0.0142}$(6)	$0.0785_{\pm 0.0120}$(21)	$0.9627_{\pm 0.0060}$(10)	$0.5527_{\pm 0.0108}$(6)	$0.4802_{\pm 0.1330}$(8)
DIF	$0.7236_{\pm 0.0113}$(9)	$0.2561_{\pm 0.0457}$(8)	$0.9701_{\pm 0.0083}$(7)	$0.4606_{\pm 0.0187}$(10)	$0.6147_{\pm 0.1393}$(5)
DiffInt-DA	$0.7813_{\pm 0.0090}$(3)	$0.2618_{\pm 0.0190}$(7)	$0.9855_{\pm 0.0032}$(4)	$0.8448_{\pm 0.0362}$(4)	$0.7143_{\pm 0.0435}$(**1**)
DiffInt-IA	$0.7744_{\pm 0.0057}$(4)	$0.2902_{\pm 0.0193}$(5)	$0.9866_{\pm 0.0032}$(3)	$0.8537_{\pm 0.0371}$(3)	$0.7109_{\pm 0.0423}$(2)
DiffInt-PA	$0.8143_{\pm 0.0034}$(**1**)	$0.3010_{\pm 0.0216}$(4)	$0.9911_{\pm 0.0027}$(2)	$0.9647_{\pm 0.0053}$(**1**)	$0.7004_{\pm 0.0484}$(3)
ECOD	$0.5576_{\pm 0.0058}$(20)	$0.4516_{\pm 0.0447}$(**1**)	$0.9559_{\pm 0.0078}$(13)	$0.1988_{\pm 0.0010}$(20)	$0.5990_{\pm 0.1003}$(6)
GMM	$0.7397_{\pm 0.0068}$(7)	$0.2793_{\pm 0.0472}$(6)	$0.9404_{\pm 0.0117}$(15)	$0.4953_{\pm 0.0018}$(7)	$0.3155_{\pm 0.1184}$(13)
HBOS	$0.6460_{\pm 0.0072}$(16)	$0.2150_{\pm 0.0363}$(12)	$0.9668_{\pm 0.0108}$(8)	$0.3693_{\pm 0.0026}$(13)	$0.1901_{\pm 0.1312}$(14)
IForest	$0.6996_{\pm 0.0167}$(12)	$0.2438_{\pm 0.0504}$(11)	$0.9765_{\pm 0.0090}$(6)	$0.4900_{\pm 0.0084}$(8)	$0.0054_{\pm 0.0008}$(21)
KDE	$0.6595_{\pm 0.0057}$(13)	$0.1058_{\pm 0.0136}$(17)	$0.9596_{\pm 0.0069}$(12)	$0.2953_{\pm 0.0015}$(15)	$0.1038_{\pm 0.0424}$(17)
KPCA	$0.4320_{\pm 0.0090}$(22)	$0.0744_{\pm 0.0108}$(22)	$0.4836_{\pm 0.0176}$(20)	-	-
LMDD	$0.3822_{\pm 0.0185}$(23)	$0.0937_{\pm 0.0280}$(18)	$0.1166_{\pm 0.0400}$(22)	$0.1890_{\pm 0.0482}$(22)	$0.0717_{\pm 0.0759}$(18)
LOF	$0.5361_{\pm 0.0044}$(21)	$0.1153_{\pm 0.0104}$(15)	$0.0921_{\pm 0.0057}$(23)	$0.2298_{\pm 0.0015}$(19)	$0.1215_{\pm 0.0900}$(16)
LUNAR	$0.8041_{\pm 0.0064}$(2)	$0.3489_{\pm 0.0518}$(3)	$0.9939_{\pm 0.0074}$(**1**)	$0.9477_{\pm 0.0082}$(2)	$0.6708_{\pm 0.0870}$(4)
MCD	$0.6573_{\pm 0.0060}$(14)	$0.1425_{\pm 0.1684}$(14)	$0.8408_{\pm 0.0123}$(18)	$0.4762_{\pm 0.0021}$(9)	$0.0059_{\pm 0.0012}$(20)
MO_GAAL	$0.2282_{\pm 0.0055}$(25)	$0.0128_{\pm 0.0009}$(24)	$0.0371_{\pm 0.0007}$(24)	$0.1939_{\pm 0.0307}$(21)	$0.0003_{\pm 0.0001}$(24)
OCSVM	$0.6065_{\pm 0.0048}$(18)	$0.1124_{\pm 0.0148}$(16)	$0.9597_{\pm 0.0069}$(11)	$0.3467_{\pm 0.0021}$(14)	$0.1629_{\pm 0.0699}$(15)
PCA	$0.5939_{\pm 0.0065}$(19)	$0.1989_{\pm 0.0247}$(13)	$0.9164_{\pm 0.0063}$(17)	$0.1632_{\pm 0.0009}$(24)	$0.4109_{\pm 0.1316}$(11)
QMCD	$0.6329_{\pm 0.0082}$(17)	$0.0840_{\pm 0.0156}$(20)	$0.6512_{\pm 0.0358}$(19)	$0.2299_{\pm 0.0011}$(18)	$0.0222_{\pm 0.0313}$(19)
SO_GAAL	$0.2286_{\pm 0.0060}$(24)	$0.0128_{\pm 0.0009}$(25)	$0.0369_{\pm 0.0005}$(25)	$0.2632_{\pm 0.0935}$(16)	$0.0003_{\pm 0.0001}$(23)
Sampling	$0.7256_{\pm 0.0190}$(8)	$0.0840_{\pm 0.0224}$(19)	$0.9660_{\pm 0.0072}$(9)	$0.5764_{\pm 0.0818}$(5)	$0.4462_{\pm 0.1391}$(9)
TCCM	$0.7154_{\pm 0.0256}$(10)	$0.2440_{\pm 0.0541}$(10)	$0.9840_{\pm 0.0041}$(5)	$0.4232_{\pm 0.2331}$(11)	$0.5273_{\pm 0.1615}$(7)
VAE	$0.6508_{\pm 0.0137}$(15)	$0.2533_{\pm 0.0474}$(9)	$0.9336_{\pm 0.0114}$(16)	$0.2381_{\pm 0.0014}$(17)	$0.4244_{\pm 0.1402}$(10)

TABLE XXVII: AUPRC, high-dimensional datasets (1/2).

Model	3_backdoor	5_campaign	9_census	17_InternetAds	24_mnist	25_musk
ABOD	$0.0506_{\pm 0.0021}$(19)	$0.1690_{\pm 0.0041}$(19)	$0.0467_{\pm 0.0004}$(13)	$0.2707_{\pm 0.0321}$(19)	$0.2124_{\pm 0.0143}$(17)	$0.0349_{\pm 0.0173}$(20)
AutoEncoder	$0.8526_{\pm 0.0112}$(6)	$0.3426_{\pm 0.0116}$(8)	-	$0.8064_{\pm 0.0288}$(2)	$0.5906_{\pm 0.0384}$(4)	$1.0000_{\pm 0.0000}$(**1**)
CBLOF	$0.0859_{\pm 0.0166}$(16)	$0.2479_{\pm 0.0200}$(17)	-	$0.5940_{\pm 0.0186}$(8)	$0.4009_{\pm 0.0413}$(12)	$1.0000_{\pm 0.0000}$(**1**)
DIF	$0.7901_{\pm 0.0356}$(9)	$0.2486_{\pm 0.0192}$(16)	-	$0.2758_{\pm 0.0231}$(18)	$0.4513_{\pm 0.0344}$(11)	$0.9963_{\pm 0.0040}$(13)
DiffInt-DA	$0.8845_{\pm 0.0054}$(3)	$0.3649_{\pm 0.0065}$(4)	$0.1077_{\pm 0.0016}$(5)	$0.5959_{\pm 0.0084}$(7)	$0.6106_{\pm 0.0152}$(3)	$1.0000_{\pm 0.0000}$(**1**)
DiffInt-IA	$0.8799_{\pm 0.0036}$(4)	$0.3514_{\pm 0.0056}$(6)	$0.1037_{\pm 0.0010}$(6)	$0.4813_{\pm 0.0406}$(14)	$0.5731_{\pm 0.0240}$(6)	$1.0000_{\pm 0.0000}$(**1**)
DiffInt-PA	$0.8897_{\pm 0.0092}$(2)	$0.3698_{\pm 0.0056}$(2)	$0.0767_{\pm 0.0005}$(8)	$0.5616_{\pm 0.0068}$(11)	$0.5209_{\pm 0.0264}$(8)	$0.9998_{\pm 0.0025}$(12)
ECOD	$0.1094_{\pm 0.0036}$(15)	$0.3696_{\pm 0.0115}$(3)	-	$0.5749_{\pm 0.0143}$(10)	$0.2070_{\pm 0.0118}$(18)	$0.9606_{\pm 0.0175}$(16)
GMM	$0.8557_{\pm 0.0138}$(5)	$0.3608_{\pm 0.0071}$(5)	-	$0.8295_{\pm 0.0217}$(**1**)	$0.5760_{\pm 0.0312}$(5)	$1.0000_{\pm 0.0000}$(**1**)
HBOS	$0.0380_{\pm 0.0011}$(21)	$0.3864_{\pm 0.0133}$(**1**)	-	$0.2541_{\pm 0.0652}$(20)	$0.1543_{\pm 0.0124}$(20)	$1.0000_{\pm 0.0000}$(**1**)
IForest	$0.0522_{\pm 0.0083}$(17)	$0.3123_{\pm 0.0210}$(12)	$0.0746_{\pm 0.0029}$(9)	$0.1604_{\pm 0.0224}$(25)	$0.3811_{\pm 0.0537}$(13)	$0.3308_{\pm 0.1853}$(18)
KDE	$0.7243_{\pm 0.0222}$(10)	$0.2834_{\pm 0.0111}$(15)	-	$0.7507_{\pm 0.0180}$(5)	$0.6664_{\pm 0.0270}$(2)	$1.0000_{\pm 0.0000}$(**1**)
KPCA	-	$0.0878_{\pm 0.0021}$(25)	-	$0.2184_{\pm 0.0346}$(22)	$0.0939_{\pm 0.0064}$(23)	$0.0376_{\pm 0.0149}$(19)
LMDD	-	$0.1092_{\pm 0.0043}$(22)	-	$0.4432_{\pm 0.0183}$(15)	$0.0856_{\pm 0.0057}$(25)	$0.0299_{\pm 0.0031}$(21)
LOF	$0.2406_{\pm 0.0143}$(11)	$0.1358_{\pm 0.0030}$(20)	$0.0613_{\pm 0.0005}$(11)	$0.4235_{\pm 0.0355}$(16)	$0.2629_{\pm 0.0225}$(16)	$0.0259_{\pm 0.0051}$(22)
LUNAR	$0.8908_{\pm 0.0123}$(**1**)	$0.2889_{\pm 0.0122}$(14)	-	$0.8010_{\pm 0.0240}$(4)	$0.6665_{\pm 0.0287}$(**1**)	$1.0000_{\pm 0.0000}$(**1**)
MCD	$0.2331_{\pm 0.1673}$(12)	$0.3155_{\pm 0.0121}$(11)	$0.1879_{\pm 0.0548}$(**1**)	$0.2296_{\pm 0.0375}$(21)	$0.3740_{\pm 0.1058}$(14)	$0.9951_{\pm 0.0116}$(14)
MO_GAAL	$0.0519_{\pm 0.0835}$(18)	$0.0998_{\pm 0.0183}$(24)	-	$0.1865_{\pm 0.0175}$(24)	$0.1068_{\pm 0.0420}$(22)	$0.0199_{\pm 0.0031}$(24)
OCSVM	$0.1114_{\pm 0.0038}$(14)	$0.3116_{\pm 0.0120}$(13)	-	$0.5813_{\pm 0.0144}$(9)	$0.4554_{\pm 0.0251}$(10)	$0.8724_{\pm 0.0631}$(17)
PCA	$0.0246_{\pm 0.0007}$(22)	$0.1134_{\pm 0.0017}$(21)	$0.0617_{\pm 0.0005}$(10)	$0.2987_{\pm 0.0112}$(17)	$0.0922_{\pm 0.0043}$(24)	$0.9999_{\pm 0.0004}$(11)
QMCD	$0.0128_{\pm 0.0004}$(23)	$0.3406_{\pm 0.0152}$(9)	$0.1194_{\pm 0.0013}$(3)	$0.7418_{\pm 0.0328}$(6)	$0.1640_{\pm 0.0134}$(19)	$0.0159_{\pm 0.0016}$(25)
SO_GAAL	$0.0443_{\pm 0.0635}$(20)	$0.1026_{\pm 0.0209}$(23)	$0.0600_{\pm 0.0038}$(12)	$0.1898_{\pm 0.0100}$(23)	$0.1099_{\pm 0.0404}$(21)	$0.0203_{\pm 0.0033}$(23)
Sampling	$0.1406_{\pm 0.1835}$(13)	$0.2230_{\pm 0.0340}$(18)	-	$0.5368_{\pm 0.0269}$(13)	$0.3388_{\pm 0.1105}$(15)	$0.9913_{\pm 0.0105}$(15)
TCCM	$0.8419_{\pm 0.0155}$(8)	$0.3291_{\pm 0.0169}$(10)	$0.0950_{\pm 0.0033}$(7)	$0.5548_{\pm 0.0152}$(12)	$0.5063_{\pm 0.0472}$(9)	$1.0000_{\pm 0.0000}$(**1**)
VAE	$0.8466_{\pm 0.0085}$(7)	$0.3510_{\pm 0.0091}$(7)	$0.1119_{\pm 0.0020}$(4)	$0.8043_{\pm 0.0291}$(3)	$0.5727_{\pm 0.0339}$(7)	$1.0000_{\pm 0.0000}$(**1**)

TABLE XXVIII: AUPRC, high-dimensional datasets (2/2).

Model	26_optdigits	35_SpamBase	36_speech	arrhythmia
ABOD	$0.0562_{\pm 0.0392}$(14)	$0.3895_{\pm 0.0078}$(22)	$0.0247_{\pm 0.0103}$(9)	$0.3791_{\pm 0.0363}$(20)
AutoEncoder	$0.1018_{\pm 0.0108}$(8)	$0.7571_{\pm 0.0189}$(2)	$0.0220_{\pm 0.0086}$(14)	$0.4081_{\pm 0.0651}$(17)
CBLOF	$0.1253_{\pm 0.0305}$(7)	$0.4588_{\pm 0.0198}$(14)	$0.0217_{\pm 0.0084}$(16)	$0.4896_{\pm 0.0815}$(10)
DIF	$0.0567_{\pm 0.0228}$(13)	$0.4532_{\pm 0.0279}$(16)	$0.0161_{\pm 0.0048}$(22)	$0.4995_{\pm 0.0716}$(9)
DiffInt-DA	$0.3809_{\pm 0.0592}$(3)	$0.5819_{\pm 0.0138}$(13)	$0.0213_{\pm 0.0015}$(19)	$0.5583_{\pm 0.0411}$(2)
DiffInt-IA	$0.0964_{\pm 0.0151}$(11)	$0.6032_{\pm 0.0102}$(11)	$0.0312_{\pm 0.0056}$(3)	$0.5721_{\pm 0.0361}$(**1**)
DiffInt-PA	$0.3739_{\pm 0.0616}$(4)	$0.5853_{\pm 0.0083}$(12)	$0.0233_{\pm 0.0046}$(11)	$0.5067_{\pm 0.0406}$(7)
ECOD	$0.0385_{\pm 0.0051}$(19)	$0.6205_{\pm 0.0105}$(10)	$0.0227_{\pm 0.0094}$(12)	$0.5172_{\pm 0.0701}$(5)
GMM	$0.0719_{\pm 0.0093}$(12)	$0.7368_{\pm 0.0186}$(5)	$0.0329_{\pm 0.0182}$(2)	$0.3955_{\pm 0.0553}$(19)
HBOS	$0.1829_{\pm 0.0261}$(6)	$0.7049_{\pm 0.0365}$(8)	$0.0250_{\pm 0.0114}$(6)	$0.5092_{\pm 0.0746}$(6)
IForest	$0.0992_{\pm 0.0195}$(10)	$0.7811_{\pm 0.0239}$(**1**)	$0.0244_{\pm 0.0150}$(10)	$0.5250_{\pm 0.0838}$(4)
KDE	$0.8656_{\pm 0.0478}$(**1**)	$0.4445_{\pm 0.0127}$(19)	$0.0465_{\pm 0.0207}$(**1**)	$0.5008_{\pm 0.0690}$(8)
KPCA	$0.0317_{\pm 0.0060}$(20)	$0.4376_{\pm 0.0188}$(20)	$0.0197_{\pm 0.0073}$(21)	$0.1603_{\pm 0.0393}$(23)
LMDD	$0.0273_{\pm 0.0036}$(22)	$0.4499_{\pm 0.0337}$(18)	$0.0199_{\pm 0.0056}$(20)	$0.2404_{\pm 0.0510}$(21)
LOF	$0.0524_{\pm 0.0200}$(15)	$0.3528_{\pm 0.0058}$(23)	$0.0249_{\pm 0.0111}$(7)	$0.4532_{\pm 0.0668}$(14)
LUNAR	$0.8328_{\pm 0.0875}$(2)	$0.7526_{\pm 0.0346}$(3)	$0.0311_{\pm 0.0148}$(4)	$0.5266_{\pm 0.0686}$(3)
MCD	$0.0388_{\pm 0.0063}$(18)	$0.7187_{\pm 0.0498}$(7)	$0.0248_{\pm 0.0108}$(8)	$0.4173_{\pm 0.0752}$(16)
MO_GAAL	$0.0237_{\pm 0.0072}$(24)	$0.3057_{\pm 0.0269}$(24)	$0.0143_{\pm 0.0030}$(24)	$0.1156_{\pm 0.0318}$(24)
OCSVM	$0.0401_{\pm 0.0056}$(17)	$0.4521_{\pm 0.0128}$(17)	$0.0218_{\pm 0.0088}$(15)	$0.4846_{\pm 0.0678}$(12)
PCA	$0.0298_{\pm 0.0043}$(21)	$0.4149_{\pm 0.0104}$(21)	$0.0216_{\pm 0.0085}$(18)	$0.4433_{\pm 0.0536}$(15)
QMCD	$0.0163_{\pm 0.0024}$(25)	$0.6210_{\pm 0.0247}$(9)	$0.0133_{\pm 0.0024}$(25)	$0.2007_{\pm 0.0700}$(22)
SO_GAAL	$0.0240_{\pm 0.0080}$(23)	$0.3042_{\pm 0.0237}$(25)	$0.0144_{\pm 0.0036}$(23)	$0.1119_{\pm 0.0279}$(25)
Sampling	$0.0996_{\pm 0.0674}$(9)	$0.4545_{\pm 0.0162}$(15)	$0.0225_{\pm 0.0109}$(13)	$0.4849_{\pm 0.0726}$(11)
TCCM	$0.2340_{\pm 0.1204}$(5)	$0.7235_{\pm 0.0224}$(6)	$0.0284_{\pm 0.0127}$(5)	$0.4658_{\pm 0.0748}$(13)
VAE	$0.0403_{\pm 0.0077}$(16)	$0.7460_{\pm 0.0198}$(4)	$0.0216_{\pm 0.0085}$(17)	$0.4004_{\pm 0.0621}$(18)